QUESTION FOR DISCUSSION AND RECAPITULATIONS

1. ExplainLecture Method and their advantage and disadvantage.
2. Explain about Teacher centered methods of commerce and accountancy.
3. Explain about Learner centered methods of commerce and accountancy.
4. Explain Lecture –Demonstration Method and their advantage and disadvantage.
5. Describe Characteristics, Types and steps of team teaching and their advantage and disadvantage.
6. Describe Self-Learning.
7. Explain Forms of Self Learning: Programmed Instruction/ Programmed Learning Stage.
8. Explain Types of Programming.
9. Explain Computer Assisted Instruction (CAI).
10. Spell out the Keller Plan - Personalized System ofInstruction
11. Explain the Project Method.
12. Spell out Activity Based Learning (ABL).
13. Spell out Active Learning Method (ALM).
14. Spell out Advanced Active Learning.
15. Explain the Mind Maps and Concept Maps.
16. Difference between Concept Map and Mind Map.

UNIT 5

INSTRUCTIONAL MEDIA

5.1 NATURE OF INSTRUCTIONAL MEDIA

Media are used to help the learner achieve the learning objectives more effectively and efficiently. You are familiar with common instructional media (generally called 'teaching-learning aids') such as blackboard (chalkboard), charts, modek, video film, radio, etc. Some of these media are used to create readiness in the learners for learning experiences. They provide clarity, precision and accuracy in processing information. They create visual images which help retention of the learnt concepts. Some of them also provide stimulation to more senses than one (e.g. video film or instructional television). Many media have the capacity to provide real (direct) or almost real experiences such as cardboard models of geometrical concepts, cuboids or case studies for learning principles of management. Some media provide opportunity to the learner to learn individually at his/her own pace (e.g. computer-assisted instructional programme) or in a small group (models, assignments, newspaper cuttings for discussion, etc.) or in a large (e.g. 35 m.m. film or slides). Instructional media can be used in all phase's viz. development, organisation and summarizing phases of classroom instruction. They can be used with learners of any age.

5.2 CLASSIFICATION OF INSTRUCTIONAL MEDIA

Instructional media can be classified on the basis of their characteristics. The nature and major characteristics of media. Let us now discuss how to classify these media using their important characteristics. We will use four characteristics for this purpose:

- Stimulation provided to sense organs,
- learner's control over media,
- Type of experience they provide, and o their reach.

5.2.1 Classification of Media according to the Sense they stimulate

Box 1 lists media in a traditional classification i.e. according to the senses they stimulate and the message code they transmit. It may be noted that new emerging media (see item viii) box I) stimulate more than one sense; they stimulate not just the ear or eye but both and sometimes touch. These media function in a more interactive way.

Audio

- Voice (any human sender of the message)
- Gramophone records
- Audio tapes, to be used in a tape recorder or language laboratory
- Stereo records/tapes
- Radio
- Telephonic conversation

Visual (verbal) print or duplicated

- Textbooks, supplementary books
- *Reference books, encyclopedia, etc.*
- Magazines, newspapers, etc.
- Documents, clippings from published material
- Duplicated written material

Visual (non-projected, two-dimensional)

- Messages/pictures on roll-up board
- Flat picture, cut-outs
- Posters, charts, graphs, etc.
- Cartoons, comics, etc.

Visual (non-projected, three-dimensional)

- *Models, mock-ups, display materials*
- Diagrams
- Globes or maps (three-dimensional)

- Specimens (animate or inanimate)
- Puppets

Visual (projected-still)

- Slides
- Filmstrips
- Overhead transparencies
- Micro image system: microfilm, micro card,

Audio-visual (projected-motion)

- *Film*
- Television
- Close-circuit television
- Video cassettes

Multimedia packages (for more than one sense)

- Slide + tape
- Slide + tape + workbook
- Radio + slide or posters (radio vision)
- Film + posters -t- workbook (print materials)
- Television + workbook (print materials)
- *Any of the above + introductory and summarizing talk by the teacher/*
- leader of the group
- New emerging media (all of these are multisensory)
- Teleconferencing (Group discussion through telephones)
- Cable television (localized television where feedback is possible)
- Television/communication satellites
- Computer networking o Video discs
- Mini computers/ microcomputers /word processors.

5.2.2 Classification of Media according to the Learner's Control

Another characteristic of media which can be used to classify them is the extent to which they can be controlled by the learner.

While using a textbook or an audio tape for learning, the learner can use them at his/her own pace, he/she may go back and read the paragraph or listen to the part of the programme again and again. So these may be called learner-controlled media.

Now think of a television programme. The messages are transmitted according to the pace of the seader and it may not be appropriate for an individual learner. Mass media like radio or TV offer no control to the learner.

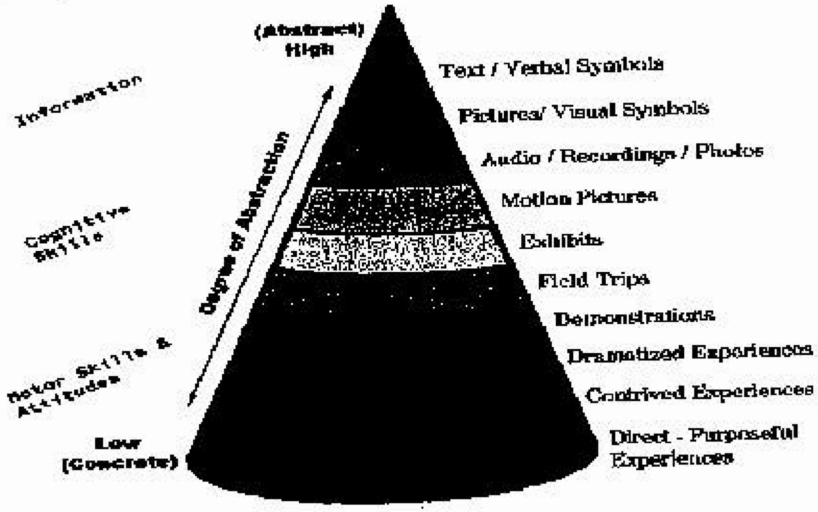
Thus various media can be placed on a continuum ranging from no control by the user to high control by the user.

1	2	3	4
No control (TV, Radio)	Projected Media	Non-projected Media	High control (Computer, Tape recorder)

All new emerging media are designed to provide more and more control to the learner over the learning process.

5.2.3 Classification of Media according to the Type of Experience they provide

Media can be classified on the basis of the type of experience they provide. The cone of experience has a broad base of direct purposeful experience which can be provided mostly through instructional media consisting of real objects, specimens and 56 methods such as field visits, observation, experimentation, etc. As we move away from the base towards abstraction (verbal messages only), we come across many instructional media which provide indirect or vicarious experiences. They differ in their degree of abstraction. Case studies, video and TV programmes provide more life-related experiences whereas radio or audio tapes tend to provide only verbal experiences.



Graphic courtesy of Edward L. Counts, Jr.

5.2.4 Classification of Media according to their Reach

Media can also be classified according to the size of the group of learners (or an individual learner) for whom they are intended and used.

A computer-assisted instructional (CAI) programme, for example, is prepared for individualized learning. It takes into account the difficulties which may be encountered by the individual learner (who is learning on his/her own without any help from the teacher or peers) in his/her learning and it aims at providing help to solve/overcome these difficulties.

Non-projected aids or graphic aids, such as charts, maps or models are used for a small group of learners. They are not quite useful to teach a class of 70-80 students.

Projected aids, such as slides or filmstrips can reach a larger group because of their quality of enlargement. Media, such as TV, radio or newspapers (print) reach millions of people at a time. Hence they are called mass media.

We shall discuss the preparation and use of instructional media in detail in following sections. For this discussion we will classify the media into three major classes' viz. individualized media, media for a small group, and mass media.

5.3 USE OF MASS MEDIA IN CLASSROOM INSTRUCTION

5.3.1 Mass Media

We have discussed above the visual non-projected and projected media which are mainly prepared by the teacher for use in his/her own class. But he/she may also make use of mass media such as print (newspaper, etc.), radio and TV for instructional purpose. Mass media are produced on a large scale to reach the masses. Because of this characteristic, they tend to address a large audience. The teacher may not be aware of the exact profile of his/her audience. As OHP provides one-way communication, there is not any real possibility of getting feedback for the teacher (communicator). In spite of all these limitations, mass media have the capacity to reach a relatively large audience at a time, with authentic, up-to-date information. Now let us examine how a teacher can make use of mass media viz., newspaper, radio, or television for classroom instruction.

5.3.2 Newspapers

- "Social studies programs should reflect the changing nature of knowledge, fostering entirely new and highly integrated approaches to resolving issues of significance to humanity" Sunrall &Schillinger Using newspaper in the classroom is fulfilling that request and appealing Students at the same time
- It may also help the history teacher in reducing the boredom and disinterest in history
- It allows students to form a link between historical concepts and modern issues, and also helps them to become more informed citizens and active and avid readers.
- Newspaper is a rich source of information. It communicates authentic and firsthand information, e.g. earthquakes in Maharashtra or in U.P. were immediately reported by the newspapers. Photographs, recent scientific information, new strategies to cope up with natural calamities like earthquake, instructions for general public, etc. are all published by the newspapers. Contents of these news items relates to the school curriculum.
- In what ways can a teacher use the newspaper in classroom? There are several ways. A teacher may use a bulletin board to display the newspaper cuttings on a particular topic. Certainly you would like to involve your students in this project. The display material could be changed every week or so. Students can be given the task of writing daily important news items on the chalkboards in the corridor.
- Yet another important method is of preparing files of newspaper cuttings on different topics or issues. These cuttings can then be used for conducting group discussions, giving group assignments or even for individual study. These files come handy when you require information on a particular aspect of a topic. The files prepared by the teacher and students can be maintained in the school library.

Tips for Using the Newspaper in Class

- Allow students time to read the paper.
- Focus on one section at a time.

- Introduce new vocabulary words first.
- Explain the functions of a newspaper and how it works before you start a lesson.
- Use the sports section to reinforce math concepts.

ARCHIVES

- A collection of historical documents or records providing information about a place, institution, or group of people.
- Documents--diaries, letters, drawings, and memoirs--created by those who participated in or witnessed the events of the past tell us something that even the best-written article or book cannot convey.
- It exposes students to important historical concepts where they learn to evaluate evidence, acquire insight into the basis on which historical arguments are developed.
- Students become aware that all written history reflects an author's interpretation of past events.
- Students read a historical account, they can recognize its subjective nature and they develop important analytical skills.

5.3.3 Radio Talk

Radio broadcast and audio recordings are the sources of audio learning experiences for the children. In order to provide learning experience beyond the school syllabus and to relate it to the real life outside the classroom, school broadcast programmes could be one of the best medium. It may not always be possible for a Computer Science teacher to invite eminent persons of Computer Science for the lecture or talk. In such cases the lectures or speeches can be pre-recorded and can be played in the classrooms. There are various types of programmes, such as discussion forums, question-answers, debates, quizzes, and speeches, dramas, which can be either played live or can be pre-recorded, to be used in teaching learning of Computer Science.

The All India Radio has regular programmes for school children. Programmes generally include talks on educational, scientific, current topics, etc. The topic, date and time of broadcast of such talks are given in advance. The schools can take advantage of such talks. Sometimes, it is also possible to synchronise the broadcast on a topic with the actual teaching-learning time of that topic in the class.

In many states of India All India Radio (AIR) broadcast school programmes. The school radio generally broadcasts programmes based on primary and secondary school curriculum. They are broadcast in the state language. A copy of the broadcast schedule for a term/ semester and the syllabus (i.e. content covered) are sent to schools so that schools can plan their time-table accordingly.

The school broadcast can be used and are in fact being used in two ways: one, directly at the time of broadcast, and secondly by recording the programme on audio cassette and using it whenever convenient. The audio cassette technology gives more control to the users- the teacher and the student. A catalogue of such taped programmes can be made for easy retrieval and use.

Non-broadcast mode (i.e. specially produced audio cassettes) is also useful in classroom instruction. Institutions such as Central Institute of Educational Technology, New Delhi, State Institute of Educational Technology, Educational Technology Cells of State Council of Educational Research and Training produce need-based audio programmes for school children. For example, State Institute of Educational Technology, Pune has produced audio cassettes on English textbooks for grades V and VI. Teachers can use them as an instructional aid.

Audio cassettes/radio programmes are generally prepared on topics which are more suitable to verbal communication. Sound, music, and special audio effects can be used in audio programmes as to make them more effective. These techniques help create visual images through sound.

- To get the maximum benefit from such talks, the following points should be kept in mind:
- To keep students' interest alive, they are facilitated to get familiar with the background of the talk beforehand. A discussion could be arranged after the talk.
- Preferably short duration talks are arranged.
- The students having hearing problems are seated near the source.

5.3.4 Television

- The television in the present day society can be used as one of the important teaching-learning aids. It combines the advantages of a radio (broadcast) and of a film, and could be used for mass education. Topics of discussion can be

announced in advance and teacher can easily carry on teaching-learning process around the telecast time to incorporate them in the on-going lesson so that students can watch and discuss the concepts in the class. Such teaching-learning helps students to develop their interest in the subject. UGC programmes are telecast on Doordarshan. NCERT telecasts its educational programme on Gyan Darshan channel.
- There is a tremendous potential to increase television based education as we have a dedicated satellite in the geostationary orbit named EDUSAT. A large number of educational television programmes can be made and telecast. EDUSAT also offers a facility for two-way interaction where the viewers can raise their doubts and make comments also. NCERT also uses video-conferencing mode to interact and train teachers all over India. A large number of scientific programmes on scientific issues are telecast on various channels of television. Teacher herself needs to be aware of such programmes to guide her students.
- National Geographic, Discovery, Discovery Computer Science are informative channels made available through television. These channels enable us to see many programmes on scientific issues with High Definition (HD) transmission and give a better, more vivid watching experiences which are of educational value.

5.3.5 Instructional TV

In most of the states, school television telecast instructional programmes directly based on or related to syllabus. A telecast schedule may be provided to schools so as to provide time for TV viewing by that particular class. School TV programmes can also be taped on the video cassette with the help of the video tape recorder (VCR). Recorded video programmes provide more control to both the teacher and the learners.

State Institutes of Educational Technology in 'six states (viz. Maharashtra, Orissa, Karnataka, U.P., Andhra Pradesh, and Bhar) and the Central Institute of Educational Technology, New Delhi, produce educational TV programmes for young children (age , group 5 to 11 years). These programmes are produced in state languages and are telecast through a communication satellite. These programmes are not (neither they are intended to be) directly based on the syllabus of primary education, but they provide an opportunity to children to enjoy learning outside the four walls of the classroom.

The thud major instructional TV programme is UGC's Countrywide Classroom. They are produced by EMRC (Educational Media Resource Centres) and AVRC (Audio Visual Research Centres) all over the country established by UGC. Though their intended audience are college and university students and English is used as the language of communication, many of the programmes can be utilized by the teachers and students of secondary and higher secondary levels.

There are some educational programmes such as Wild America, Ascent of Man, Quest, Turniog Point, etc., on related subjects which are telecast on National Network. If possible, the relevant programmes can be recorded on video cassettes and used as and when required.

Students should be exposed to such instructional TV programmes, if not in school, then at home or using a community TV set.

One major limitation of TV is the requirement of electricity. Maintenance and repair also pose problems. Inspite of these limitations, TV is a rich source of educational inputs.

5.3.6 Integration of Media

Communication media, discussed so far, are not to be used in isolation. Mostly, they play a supportive role to various interactive methods and techniques used by the teachers. A teacher may choose to use a chalkboard, a flannel board and charts to supplement his teaching. In this situation he should integrate all these media so as to give rich learning experiences to the learners. In the same way OHP can be effectively used in a seminar. A group discussion can be initiated after showing a set of slides.

The important point to remember is that the media should not be used because they are available but they should be integrated with other techniques in such a way that the integration provides meaningful experiences to the learners.

5.4 EMERGING MEDIA

So far we have learned and discussed about the traditional media such as print and electronic media. We have also learnt about their style of functioning. The print media includes both newspaper and magazines along with brochures, pamphlets and others. As far as radio is concerned, we meant it as All India

Radio. Just a decade ago we have started listening to various private FM channels. These days we are very much used to with the news and entertainment television channels. These are otherwise known as mainstream media.

With the emergence of online technology, the production, distribution and consumption process of news has been changed. Newspapers and television channels have shifted their importance to the online editions. Each and every media houses in India now have their own websites. Especially the electronic communication with the internet driven technology has brought lots of innovations. The new forms of media have created opportunity for interactions between the producers and consumers of news. These new forms of media are known as emerging media.

Journalism is moving slowly into interactive mode. Readers are now relying on online versions to get instant information rather than on traditional media. They are also reacting and engaging with news like never before. Newspapers are not only carrying text but also uploading videos and graphics in their websites. Online communities and Blogs are made to discuss the current affairs. Journalists are posting news, features and columns in their blogs. Personal blogs are linked to the websites of media houses. On television News anchors are giving feedback to the news sources. News anchors are at times becoming news makers. They are the active commentators. Opinion journalism, collaborative journalism and crowd-sourced journalism are new concepts found in the field of journalism.

5.4.1 What Is Emerging Media?

In common parlance, by "emerging" we mean "upcoming". So in journalism, by "emerging media" we mean the upcoming media used to access information. It is very difficult to define exactly what the emerging media is meant by. As media is evolving in nature so also its various forms. World Wide Web, Internet and digital technology have influenced both print and electronic media. American Journal of Business defines that any definition of emerging media is difficult. The most commonly applied 'shorthand description' of emerging media is that it is communications – of all types – based on digital technologies, and increasingly with interactive components. Neuman (1991) has defined the concept of emerging media more than a decade ago. He argued that what we define as emerging media will

a) Alter the influence of distance,

b) Increase the volume and speed of communications,

c) Enable interactive communications and d) permit the merging of media forms.

Thus we can define emerging media as "media used to share and exchange information with the use of emergent digital technologies having the feature of interactivity, on demand and faster speed." Thus emerging media not only contain new medium of communication but should also have the faster speed of delivery of information. New media definitions remain fluid and are evolving, with some definitions of new media focusing exclusively upon computer technologies and digital content production whilst others stress the cultural forms and contexts in which technologies are used (Dewdney & Ride, 2006). In other words, emerging media is the one that employs of digital technology to communicate message to the masses in a new and innovative ways. It has altered the influence of distance. Emerging media enables interactive communication and permits the merging of various media forms. We are also seeing the convergence of media forms.

5.4.2 Types of Emerging Media

We are now living on a virtual world. We are communicating with people without knowledge on their physical presence in the virtual world. While browsing the websites we can now interact with the content of the website. You must have seen websites where you can watch live or recorded videos or audio files. There are features available now, that allows you to upload your own video in these websites. With smart phones you are communicating with friends resided in a far off place within fraction of seconds. All these are happening with digital technology enabled emerging media. Now traditional media outlets are deploying emerging media to create a new user experience. Communication with the help of emerging media is the next big thing we are going to witness. Let us know about the different types of emerging media either available or coming up in the near future.

5.4.3 Virtual Reality:

Another emerging media technology on which there are lots of discussions going around is virtual reality. Though the word seems new but it has been there since quite long. The word virtual reality is a combination of two words, 'virtual' which mean near and

reality means truth or what is actually happening around us. So it is basically a technology which is using both software and hardware components of a computer. It creates a three-dimensional environment which human beings can interact. A person can do a lot of actions and is in a position to manipulate objects with virtual reality. Virtual reality creates an artificial environment where sensory experiences can be achieved.

It is a kind of digital world where illusion of reality is created for the human senses. Video games are perfect examples of virtual reality. When we are playing racing car or bike games, it feels like that we are driving the vehicle. Virtual reality has applications in different fields such as sports, entertainment, medicine, architecture, media and others. Virtual reality has been used in films and television programmes. The Matrix and Vanilla Sky are some of the films where virtual reality has been used. Books writers, publishers and art work designers are using virtual reality to create a 3D environment to communicate with their target audience. Virtual reality has been used to explain any historical instances also. Application of virtual reality is found in the medical and architecture education. It creates a more flexible environment to work and thus helps to reduce time and cost for many organizations. Virtual reality is considered to be a prime technology behind the development of various advanced technical products such as motion tracking, movement sensors and others.

The above image depicts an eyewear available in the market to experience the virtual reality. In order to feel the virtual reality, a person need to wear a pair of special types of gloves and a head mounted display. The gloves will help to receive the computer input. Apart from these two, there are other types of tools available in the market for the users. We can say that virtual reality is a reality.

5.5 SATELLITES

Satellite is a spacecraft that receives signals from a transmitter on earth and amplifies these signals, changes the carrier frequencies, and then retransmits the amplified signals back to the receivers on earth.

The space age and launch of satellites started in 1957 with the launching of Sputnik by the former USSR. Since then, a number of satellites have been launched for various purposes like telecommunications, meteorology, remote sensing, disaster warning, defense and so on.

Till date, thousands of satellites have been launched into orbit around the Earth. These originate from more than 50 countries. Moon is a natural satellite on earth and the rest are artificial (man-made) satellites.

5.5.1 Types of Satellite

Satellites can be categorized into different types. The basic types are:

- Communication
- Weather
- Navigational
- Reconnaissance
- Application
- Research

Type of Satellite	Description
Communication Satellite	The first communication satellite was Echo 1 launched in 1960. Relay 1 and Telstar 1, both launched in 1962, were the first active communications satellites. INSAT - 4A was India's first communication satellite. Communication satellites provides worldwide linkup of radio, telephone and television. They beam signals around the world
Weather Satellite	Tiros was the first weather satellite launched in 1960 from Florida USA. These satellites provide continuous, update information about large scale atmospheric condition such as cloud cover and temperature profile. It also gives information of hurricanes and cyclones. There are two basics types of meteorological satellites. These are: Geostationary and Polar orbit. Geostationary type send weather data and pictures that cover a section of the United States, China, Japan, India, and the European Space Agency.
Navigation Satellite	Navigational satellites provide data to ships, aircrafts and submarines. The recent development is the Global Positioning System (GPS). GPS provides reliable location anywhere on or near the earth. It is maintained by the United States and any one can access it. It is a useful tool for map making, tracking and surveillance besides the military and civilian uses.

5.5.2 Characteristics of Satellites

The characteristics of all communication satellites are similar. These are:

i) **Power:** A live satellite does not require conventional power to maintain its position in space, except tiny amounts of energy necessary to correct its position occasionally. The power for receiving and transmitting signals comes from the solar batteries built into the satellite. These batteries are recharged. Solar panels, which convert sunlight into electrical energy, are used for the functioning of the satellite system.

ii) **Large coverage:** Satellite-based communication is independent of distances and serves the rural and urban, central and far flung areas simultaneously. It can cater to very widely dispersed populations at a time. This characteristic of the satellite is particularly useful for education at a distance. Space scientists claim that three satellites in the geo-synchronous altitude orbit can provide communication services to the entire earth on a full time basis, except for the Polar Regions, which are not visible from this orbit (Nicholson, 1976).

iii) **Multi-purpose uses:** Satellites can be used simultaneously for the radio, telephone, and television and data traffic. Multi-purpose satellites offer a wide variety of combinations. Besides serving communication purposes, the satellites are also used for remote sensing, such as is required in soil surveys, flood (assessment of area under water, etc.), forestry (tree resources, tree diseases, etc.), oceanography, etc.

iv) **Cost:** The initial investment in the development and launching of a satellite is very high, especially for the third world countries. A multipurpose satellite such as INSAT, needed a huge financial allocation in its fabrication and launching. But when INSAT-IB was launched and became operational, all demands for communication were met without adding new investment. On the other hand, the terrestrial system, including the microwave, needed additional infrastructure to meet the increasing information needs of a country. Expanding telecommunication infrastructure to provide communication services to different parts of the country is not always an economically rational thing to do. Because telecommunication for educational purposes cannot produce sufficient revenue to cover capital and operational costs, the costs in this case should be counted in terms of social and economic benefits, such as roads, water supply systems, schools, etc.

v) **Planning:** The implementation of satellite-based communication requires advance planning. It needs more lead time than terrestrial communication does. Therefore, the use of satellites should be linked to the overall socioeconomic and educational development of the country. Since it (satellite based communication) is closely linked with the educational development and economic growth of the country, it should have a base in long term planning.

vi) National and area specific communication: Satellite-based communication has the capability to cater to both the national and the area specific needs of a country. It can be regionalized as well which can provide area-specific service.

vii) Life of a satellite: The use of solar panels/cells describe the life of a satellite. The electrical energy output from a solar cell will decrease with age: after 8 to 10 years, the electrical output from a solar cell will decrease by about 20 per cent. The communication satellite is generally replaced after about 10 to 12 years of continuous service. For example the life span of Indian satellite- INSAT-1A and IB was seven years. Launched in August, 1983, INSAT-IB completed over 108 months of operational service in August, 1992.

5.5.3 Indian Experiments in Use of Satellites in Education

India launched its first satellite Aryabhatt in 1975. Since then number of satellites have been launched for various purposes. India has come to the center stage in the satellite technology and hence pushed the developed countries to the periphery.

Number of experiments has been conducted in various sectors. In the following section a few important experiments in the field of education are described

Satellite Communication	Terrestrial Communication
Does not require ground-base, high-power, high-tower systems.	Needs high-power, high-tower systems, (The height of the transmitter at Pitampura, Delhi, India is 235 metres)
Does not need much ground equipment; direct reception television sets can receive signals directly from the satellite.	Needs a number of transmitters for wide coverage (at present Doordarshan India has a network of 529 transmitters spread all over the country).
Being a highly-centralized system it provides more positive pacing and control over the developmental process.	Cost of equipment is low
Use of satellite technology could inspire the teachers, students, and parents for modernization because it brings the whole world together into a remote village.	At times difficult because of technical and managerial hindrances in systems coordination. Further, because of the position taken by the local authorities controlling the flow of information at the regional level, it becomes difficult at times.
Planning and implementation require more lead time.	Because of limited coverage, it has some drawbacks.
Planning and implementation require more lead time.	Needs comparatively less time.
Effective for a large country or a group of countries	Suitable for a small country or a part of the country. Needs comparatively less time.
Independent of distance.	Limited coverage area
Independent of the nature of terrain.	Difficult in mountainous and sea areas.
Can meet the increasing demand of communication without additional cost.	Communication capacity is bound to the regional system installation and requires additional cost.
More effective in meeting the overall needs of people at the national level.	Serves regional information needs better.

A satellite failure can result in the entire system being inoperative which might pose severe readjustment strains on the communication.	Systems failure would not be as disastrous as in the case of the satellite system; readjustment and repair of damage is manageable.
Needs parking spaces for the geosynchronous satellite which is becoming more and more scarce.	Does not need parking space.
Worldwide network connections via satellite can resolve the problem of unequal educational opportunities and can provide a truly world-wide sharing of educational, resources for international education.	Network is possible at the regional level only

5.5.4 Satellite Instructional Television Experiment (SITE)

SITE was the largest communication experiment in the use of satellite in support of developmental and educational programmes in modern times. The main impetus for the SITE project came from Prof. V. A. Sarabhai. In 1969, India and USA started an experiment called SIET by means of Applications Technology Satellite (ATS-6). On May 30, 1974, the Satellite was launched from Cape Carnival in USA. The telecast via this satellite began in India from August 1, 1975. Indian Space Research Organization (ISRO) with All India Radio (AIR) took the responsibility of broadcasting ETV programmes to the selected villages in six states of Andhra Pradesh, Bihar, Karnataka, Orrisa, Madhya Pradesh and Rajasthan, selected on the basis of their educational backwardness.

The experiment continued from August 1975 to July 1976. The instructional objectives of SITE were in the fields of education, agriculture, health and family planning and national integration. About 2400 Direct Reception Television Sets (DRS) deployed for SITE were located in different cultural, linguistics and agricultural regions of the country. Different socio-economic environments were also chosen for the purpose. Television broadcasts via satellite were made available for four hours a day, one and half hour in the morning and two and half hours in the evening. Morning times were utilized for broadcasting programmes for children which were enrichment programmes for

the age group 5 to 12 years; evening programmes were directed to adults.

SITE covered four different language regions but children of other regions also watched these programmes on school days. Though the programmes were meant for children, others also viewed the programmes within the school. These were not based on school syllabi but intended to provide general enrichment. Governments of each state receiving SITE programmes were responsible for electrifying the school, which housed the television receiver.

The contents of the programmes were identified by a group of educationists drawn from National Council of Education Research and Training (NCERT). It was then placed before the senior officials of Department of Education of each of the State Institutes of Educational Technology (SIET) in the respective states. ISRO produced a series of programmes in science, which aimed at developing scientific thinking. Production studio was also set up by ISRO in Bombay and its staff developed one of the educational series. Programmes were produced at three Base Production Centers: Delhi (Hindi), Cuttack (Odia) and Hyderabad (Telgu and Kannada). The format of the programmes was lecture demonstration followed by documentary, drama and discussion. Before approaching the programmes for broadcast purpose, a few prototypes were produced and pretested in different villages. The purpose of this pretesting was to test the acceptability of the programmes.

Experience during SITE period was quiet encouraging for further expansion of television service in the country. Government decided to start the SITE continuity community-viewing programme. Forty percent of the villages were provided community-viewing facility in six SITE cluster areas by setting terrestrial transmitters. This was possible because the infrastructure existed and studio facilities developed during SITE Terrestrial transmission was made available from 1977 to 1982 and educational programmes were available in the morning hours along with other programmes in the evening.

An important highlight for SITE was teacher training through multimedia. Nearly 50,000 teachers were exposed to this training in two installments. Experts planned the lessons. SITE experiment drew attention of the world. Two international teams, one sponsored by United Nations and other by Commonwealth

Broadcasting Association toured the SITE areas and gave favorable reactions.

5.5.5 Indian National Satellite (INSAT)

The SITE implemented in India in 1975-76 received great applause at national as well as international levels. This unique success with a borrowed satellite for one year encouraged India to have its own satellite. In 1977, India approved a proposal to launch a multipurpose and space communication system of her own called Indian National Satellite (INSAT).

The major objectives of INSAT were:

- To produce and transmit varied programmes designed to awaken, inform, enlighten, educate, entertain and enrich all sections of the people in different parts of the country.
- To promote alternative approaches to education for children, youth and adults.
- To stimulate interest and involvement of people in economic development.
- To stimulate interest and involvement of people in economic development.

INSAT-1A was launched in its orbit on April 10, 1982. According to Indian specifications and requirements, it was designed and fabricated by Ford Aerospace and Communications Corporation (FACC), USA. It was a joint venture of the Department of Space, Posts and Telegraphs, Meteorological Department, Ministry of Information and Broadcasting and Ministry of Education and Culture. All these government agencies used the satellite to reach the target groups, the Ministry of Education had a special commitment to use its facility for fulfilling the educational priorities.

INSAT-1A developed mechanical snags and in September 1982 it ceased to function. However, INSAT-1B was launched on August 30, 1983. INSAT project covered six of the educationally backward states as in SITE. About four thousand television sets were installed and commissioned.

These television programmes were telecast in the morning and evening. Morning transmission was devoted to school education for children in the age group of 5-8 years and 9-12 years. These programmes were not only syllabus oriented but provided broader perspectives of enriching school lessons. Local Doordarshan Kendras and Central Institute of Educational Technology (CIET),

NCERT New Delhi, produced programmes. Evening television programmes were mainly developmental and national programmes and included news, films and live telecasts.

India's space programmes took another big leap on July 24, 1993 when the multi-functional indigenously built satellite INSAT-2B blasted off into space by Arriane launch vehicle from KOUROV, French Guyana in South America. INSAT-2 satellite is a multipurpose satellite providing space services for telecommunication, metrological observations and data relay, nationwide TV broadcasting, radio and TV distribution disaster warning and distress alert sources. It has 50 percent higher communication capacity than the first generation INSAT-1 satellite.

5.5.6 Indira Gandhi National Open University

IGNOU established by an Act of Parliament in 1985 has an objective to take education to the doorsteps of the students and provide education to all, irrespective of age, region or formal qualification. It also offers need based vocational and professional academic programmes. The university has a nationwide network of study centres, which are equipped with CD and DVD players, Color TV sets, besides print materials. IGNOU has a full-fledged communication division - Electronic Media and Production Centre, which produces audio and video programmes and organizes educational broadcasts over television and radio for the benefit of students as well as the general public.

Broadcast of IGNOU's educational programmes began in May 1991 on the national network of Doordarshan thrice a week in the early morning. The broadcasts of audio programmes began in January 1992 from Bombay and Hyderabad and later from Delhi and Lucknow. The IGNOU programmes are syllabus based and cater to the learners enrolled in IGNOU programmes. These programmes supplement the self-instructional texts provided to the students of the university. These broadcasts mark a major step in the progress of IGNOU in fulfilling its educational objectives and in the country's development.

5.5.7 Gyan Darshan

Gyan Darshan (GD) is an exclusive and dedicated twenty four hour educational and developmental TV channel of India. It is a joint collaborative venture of Ministry of Human Resource Development (MHRD), Information & Broadcasting Ministry, IGNOU, UGC, CEC, NCERT, CIET, SIETs, National Institute of Open Schooling (NIOS), Department of Space and Technology, IITs, Technical Teachers Training Institute's, Department of Space, DECU, Ministries of Rural Development, Health, Labour, Environment, National Aids Control Organisation (NACO) etc. It was inaugurated on 26th January 2000. EMPC of IGNOU has been identified as the coordinating and transmission agency.

It offers interesting and informative programmes of relevance and value to specialized categories - preschool kids, primary and secondary school children, college/university students, and youth seeking career opportunities, housewives, adults and many others. The technical arrangements for Gyan Darshan is that the earth station was set up with a 7.2 m antenna with arrangements for the play of pre-taped programmes from the Earth Room Station itself while others can be viewed from Video Studio II. Microwave links to enable live relay of the channel from EMPC to Doordarshan Kendra, Delhi and to source programme from the CIET studios have been installed. The footprint of the satellite is the entire country and GD signals can be conveniently received without any special equipment anywhere. Gyan Darshan as it entered in its fourth year on January 26, 2004 went completely digital. It has expanded into a bouquet of channels namely GD-1, GD-2, GD-3 Eklavya, and GD-4 Vyas.

GD-1 is a twenty four hour channel devoted to education and distance education. The transmission for curriculum based and enrichment programmes are each of 12 hour duration. The programmes of IGNOU and CIET-NCERT including NIOS are telecast for four hours each. IIT programmes for three hours, CEC-UGC programmes for two and half-hours and one hour each for TTTI and adult education. It is also available in Ku-band.

GD-2 is devoted entirely to interactive distance education. It is a one way video and two ways audio satellite based interactive system operating on the C-Band transponder of INSAT-3C.

The third channel GD- 3 Eklavya, is the Technology channel which brings quality education to the students pursuing engineering education throughout the country. Eklavya features lectures of the courses taught at the IITs situated at Kharagpur, Mumbai, Kanpur, Delhi, Guwahati, Roorkee and Chennai. Eklavya transmits 24 hours daily, with eight courses running in parallel. These are repeated once for the benefit of those who may have missed viewing the first time. This pattern continues from

Monday to Saturday. Sundays are reserved for special interest programmes on Technology and Science. Eklavya – Technology channel reaches every corner of the country through INSAT 3C Satellite on C Band (74 degrees East), Downlink frequency 4165 MHz.

Another channel in the bouquet of Gyan Darshan channels is GD-4 Vyas which brings quality education to the students pursuing higher education throughout the country. Higher education in India has expanded significantly. Every district and small town of the country has colleges providing higher education to people. Today those enrolled for higher education account for 9 million students in nearly 13000 colleges and 234 institutions of higher education.

The aim of this channel is to bridge the knowledge and information gap in the area of higher education and provide information to all those who need it. The vision of Vyas therefore, is to reach out to large number of students, teachers and general public with quality educational material electronically so as to address the issues of access and equity with quality higher education.

This channel chooses those subjects, which are in high demand but lack adequate number of competent teachers. It also selects such subjects which are difficult to be explained optimally by conventional classroom tools but could be effectively covered in the visual multi-media animated form for competitive examinations, women and general public.

The primary target audiences of the Channel are the students studying in undergraduate and postgraduate classes in universities and colleges all over the country, particularly in small towns. Students pursuing correspondence courses, teachers teaching undergraduate and postgraduate courses and also the staff of training colleges, and students appearing for various competitive examinations watch this channel. The channel acts as a tool for the audience to take up fresh initiative in broadening their horizon especially in the field of career enhancement. The Vyas programmes are telecast round the clock on GD-4 and some are relayed on GD-1.

5.5.8 EDUSAT

EDUSAT- the dedicated satellite for education in India was launched on 20th September 2004 by ISRO. The satellite was launched from Sriharikota. It is the first Indian satellite exclusively built for the use of education sector. EduSat has a life of seven years in space during which it will help educational institutions to provide quality education.

The satellite is capable of providing high bandwidth two-way interaction by creating a private network of Satellite Interactive Terminals (SITs) and Receive Only Terminals (ROTs) installed all over the country. The interaction mode is based on the popular Hyper Text Transfer Protocol (HTTP) used in the Internet and web applications. Thus, the satellite enables us to create a network through which we can share existing resources (often called as digital repositories), in text, graphics, audio, and video formats; and also can create real-time interactive virtual classrooms (often called synchronous e-learning) across the country. With both these possibilities, the potential is enormous for the educational development of this country. The satellite has five Ku-band transponder providing spot beam, one Ku-band transponder providing national beam, and six extended C-band transponders covering regional beam. All put together covers the entire country through national and regional beams. Some of the major institutions using the Edusat network are: Indira Gandhi National Open University, National Council for Educational Research and Training, Consortium for Educational Communication, Visvesvaraya Technological University, Vigyan Prasar. Using the satellite communication, it has become possible to interact in real-time mode with students though two-way video and two-way audio system.

5.6 RELATIONSHIP BETWEEN MAINSTREAM MEDIA AND EMERGING MEDIA

In the present age media is considered as the most influential aspects of our lives. As students of journalism, you must be reading newspapers and watching news on television. When 24/7 news channels came, people were of the opinion that the newspapers would die down. Even after facing tough competition from television channels as well as from fellow newspapers, the print media had been able to survive. The print media has not only survived but also transformed and innovate new ideas to cater to its audiences. It has also been able to serve the society with apt content. Newspapers have brought innovations in the way of introduction of color, changes in page design and others. In the same way, when internet came, some critics pointed out that

people can now read news from websites. Therefore news channels are going to face the heat of declining viewership. Simple news dissemination is not enough for the viewers or readers. They are expecting to choose what they want to read or know, they want to contribute and form their opinion in the society too. This is not the death of mainstream journalism rather this has gave an impetus to the mainstream media to innovate themselves to provide better services. Therefore the mainstream media houses have started integrating technology into their existing system. They also want to remain abreast from their competitors.

All newspapers now have their online publications. The television channels are also not lagging behind. They have started uploading videos as well as text messages in their websites. Everyone wants to be present online. We now prefer to watch news on our phone instead of turning on television. Phones are now much nearer to us than a television set. The mainstream media organizations are utilizing every forms of emerging media to attract the attention of the users. You must have seen that television channels are using animated graphics to explain an event clearly. Though both the mainstream media and emerging media are working closely but they have certain kind of similarities and differences. Let us have a look on the similarities and differences existed between them.

5.7 SIMILARITIES BETWEEN THE MAINSTREAM MEDIA AND EMERGING MEDIA:

The primary function of the media is to inform, entertain and educate. The age old print media industry has been able to meet the ever growing demands of the readers. Newspapers and magazines of black and white era to the present forms of glossy and colorful papers have been able to provide all kinds of news to the masses. From black and white television news broadcasting to the present age of colorful channels with 24/7 news are also catering to the information hungry audiences. Be it the traditional media or emerging or new media, their primary job is to inform the society about the happenings. Both the forms target the masses.

More and more numbers of users are accepting emerging media technologies. They are joining the bandwagon either to interact or to collaborate or to contribute or simply to access information. Every form of media is trying to get the attention. The emerging media just like mainstream media informs, entertains the masses and persuades the masses. They are also critical to the development of many areas. Mainstream media are popular for their predictable form and content. Both the forms of media target the masses.

5.8 DIFFERENCES BETWEEN MAINSTREAM MEDIA AND EMERGING MEDIA:

Emerging media has become an inseparable part of human being in the recent time. The mainstream media is dependent on the emerging media to keep abreast of the competition in offering services. Mainstream media is considered as the most trustworthy and reliable. We can trust on a newspaper or a television channel while accessing news. But we can't rely on a blog or a website and raise questions on its credibility while reading news or accessing information. Emerging media has an edge over mainstream media for its flexibility and accessibility.

If we want to make a 3D animated image or video we can simply download the free software available on the internet to a desktop and can prepare it. Mass media has been able to provide content what the majority of the population want to hear or read. Therefore it is known as the mainstream media. It has also been able to form public opinion from time to time on specific issues

Due to its capability to create and distribute the content easily, the emerging media is becoming more popular than the mainstream media. There are easier channels available before the audiences to get the news either through their smart phone, tab or iPod. Digital technology has been considered as the major factor in influencing the emerging media. Therefore mainstream media houses are integrating digital technologies into their process.

Most of the forms of emerging media are available free of cost. But the gateway to opt for emerging media depends on internet. The speed of internet and the device to access internet are very important. The internet speed is a major concern in India.

Whether we want to see a streaming video or want to listen live concert or want to design a model we should have a faster internet connection. The emerging mass media functions in the same way as mainstream media functions but takes the help of digital technology. Digital devices play major role in discharging the functions.

Sr. No.	Mainstream Media	Emerging Media
1.	Newspapers and television channels are highly credible and we can trust them.	Various forms of emerging media are less trust worthy or everyone raises questions over the credibility of news published on websites.
2.	Television or newspaper journalists or contributors can't interact with their audiences or readers. Mainstream media organizations are not getting instant feedback from their users.	Content providers in case of emerging media can interact with their target audience instantly. Instant feedback is an important feature of this media.
3.	The readers or viewers can't get old news easily. They have to go to any archives or library to get old news.	It is easier for the viewers to search old news and videos with just a click of a button.
4.	News is presented to the reader in a conventional way.	News is presented to the viewer in an interactive manner.
5.	The reach of newspapers and television channels is restricted or limited to certain geographical area	The reach of this media is not limited to any geographic location.
6.	There is no flexibility in getting news in case of newspaper. But with 24/7 news channels people are getting little flexibility in accessing news.	This medium is more flexible and easy to get information either through text, video or voice at any point of time.
7.	The accessibility of news is not so easy. There is no such facility to guide users to access information.	The users are guided in every step to access information. Be it watching an interactive video or podcast, even a small child can access it.
8.	Information once published on newspapers or broadcast on television can't be changed.	Content in any stage can be changed, if any error noticed. Even after published you can edit and repost it.
9.	Messages generated and produced by the mainstream media are costlier	Here content generation and distribution processes are cheaper.
10.	Mainstream media have limited options to tell a story to the target audiences. (e.g. text, image, visuals and sound)	Emerging media uses several forms to tell a story to the audiences. (e.g. video, animation, text, visuals, images,

5.9 BENEFITS OF EMERGING MEDIA FOR JOURNALISTS

With the increasing usage of internet, media houses have also transformed their process of functioning into digital platform. Journalists are using various digital tools to present reports. The digital technology is offering rich visual experience in multiple platforms. The multiple benefits of using emerging media have attracted the attention of journalists throughout the globe. Following are some of the benefits journalists are getting from emerging media.

a) Maintaining relationship with source

Interactivity characteristic of emerging media is helping the journalists to build and maintain relationship with the sources. In the mainstream media, reporters don't have any scope to interact with the readers or audiences. The only way to get feedback from the readers in case of print media is the letters to the editor. In social media, content users are giving instant feedback to the published reports. Not only journalists but also their management is asking them to active on digital platforms and to extend relationship with the user community.

b) To expand the brand

In the highly competitive industry, media owners are in constant effort to popularize their brand among the users. In order to expand their brand they are taking attempts such as online presence of their outlet, creation of blogs by their journalists and other. Even the reporters and editors are publishing exclusive content on their blogs to create separate image. With the emerging media in place local media brands are gaining popularity across the globe. Emerging media is providing opportunities for the media outlets to expand their reach and popularity.

c) New ways in publishing story

Unlike mainstream journalists, here journalists are getting ample scope to file their stories as and when and can publish it even from their smart phone. They don't have to go to office to publish a story. Even television journalists can cover an event and file the story by sending visuals through their smart phones. Smart phones have many applications which are helping them even to edit their stories. Unlike space and time crunch in mainstream media, here writers have flexibility in extending and contracting their write up.

d) Crowdsourcing

We are seeing the trend that journalists are now logging into social networking sites to get story ideas. To make information rich reports they are collecting ample materials on their stories from various sources. Crowdsourcing means to collect information from various sources via internet. For writing features or trend stories most of the journalists are researching and getting inputs from various websites. To provide a unique experience, they are using multiple platforms such as visuals, text, images, graphics and animations. Increasing number of animation tools are used in various stories.

e) Knowledge of multiple platforms

Another important benefit journalists are getting from emerging media is that the knowledge of multiple platforms. With the convergence of technologies, contents are created for multiple platforms targeting every kind of users. A print journalist has knowledge on how to edit a video or how to create an animated graphics. In the same manner television and online journalists have knowledge on other tools. The journalists are also making themselves updated on various technologies. As technology is changing fast, so are also its usages. Journalists are also learning their hands on experience on the upcoming technology. Earlier a print journalist didn't have idea on the working of printing machine. But now journalists have knowledge on how the content is uploaded on the website, use of multimedia tools and use of interactive elements etc.

f) Instant feedback to stories

Another important benefit for the journalists is that they are getting instant feedback for their stories. Under mainstream media journalists have to wait for days or weeks or months to get feedback for their stories. But here within seconds of publishing content they are getting feedback or comments on their stories. Sometimes the instant feedback comes across the border which creates enthusiastic among journalists.

5.10 TELECONFERENCE

Teleconference in simple words, is a meeting between one individual with many people through telephone or network connection located in same city or in different cities, states and countries. It is a real time exchange of information between people. People can conduct meetings, training, demonstration, data transfer etc. Educational teleconferencing is a valuable medium for distance education. It involves the use of several media and permits interactive group communication by means of a two-way broadcast. Today, dependence on this technology is accelerating. The interest in using audio and related visual data that can be transmitted over regular telephone lines appears to be related to the growth of distance education.

5.10.1 Technical Description

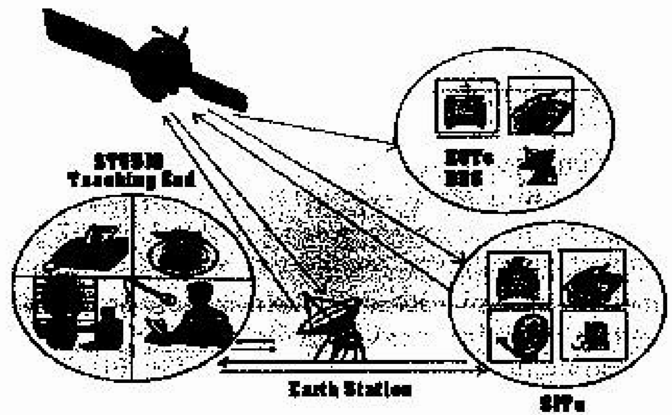
Teleconferencing is an electronic means which can bring together three or four people in two or more locations to discuss or share the use of two-way and one way video, both full motion and slow scan, electronic blackboards, facsimile, computer graphics, radio satellite and videotext. However, the most essential part of all forms of teleconferencing is a good quality audio system to help immediate interaction among the participants for information exchange.

Audio teleconferencing requires a multi-telephone line electronic switch or interconnection device called a 'bridge' to which the user can attach a wide variety of data transmission devices and telephones. The normal practice is to connect one device per line to the bridge. Audio equipments used with the bridge are the usual handsets, headsets, speaker-phones, radiotelephones, and microphone speaker units. Audio teleconferencing uses regular telephone lines provided by local public telephone companies. If the quality of standard business or household lines is good enough, virtually any line could be used. Effective contact through teleconferencing can be made in any reasonably quiet environment. Occasional users normally purchase their service from the local telephone company, teleconferencing consultants or a major user that is willing to sell available time on its system. The costs for starting a university or college-based private audio teleconferencing system are not large if the local telephone system has:

- A relatively quiet line,
- Ready accessibility, and
- Acceptable local and long distance rates.

A typical teleconference set up will have a studio with recording and transmission facility, where expert/teacher can sit and deliver lecture. This studio is linked to a satellite earth station through which live transmission goes to the satellite for further

transmission to the reception centres, where Direct Reception System (DRS) units are placed with a parabolic antenna, television set and telephone for talk-back. This is typical scenario for one way video and two-way audio teleconference system. However, in the EduSat based teleconference such centres are called Receive Only Terminals (ROTs), which may also have computers, with web-camera connected to interact with the resource persons in the studio through video as well. In addition, the EduSat network also has two-way video conference facility with Satellite Interactive Terminals (SITs) making is fully interactive in real-time mode. A typical teleconference set up is shown in Figure



5.10.2 Advantages of Teleconferencing

The support for teleconference was mainly due to the consideration of the following advantages:

Effective Support for Remote learners: Teleconferencing can be very useful when most of the potential students are widely scattered among communities that are far apart and when each center has a thin learner population, say less than ten students, in a given course or programme.

Cost effectiveness: The cost for starting and operating teleconferencing system is relatively low in comparison with other available methods of serving remote learners.

Flexible system: The system used can be adjusted quickly to serve large or small groups.

Familiar instructional mode: The mode of instruction is similar to that of the seminar with the instructor being in charge of the discussion and able to stimulate multi-location interaction.

Easy scheduling adjustments: A scheduling adjustment can be made almost as readily as for the on-campus classrooms.

Multi-locational access-control: Access to the instruction in the programme can be controlled through a limited number of off-campus centers.

High-quality instruction: The quality of instructional materials can be kept high because of the need for careful and early preparation.

Immediate feedback: The teleconferencing system provides the facility for immediate feedback to the learners and allows them to convey their reactions to the tutors.

5.10.3 Limitations of Teleconferencing

Teleconferencing has certain inherent limitations, due to which it is not in frequent use in education. Some of these are as follows:

i) Teleconferencing requires a huge and very efficient telephonic, radio and television network throughout the country.

ii) The chances of technical breakdown are quite high.

iii) The telephone charges are very high, which all the educational institutions cannot afford.

iv) Teleconferencing is a costly technique of instruction. It requires sophisticated technology and expert human power.

v) Teleconferencing is a mode of group communication, so the willingness of each participant is an essential requirement, but this is generally lacking especially among distance learners

vi) It takes time to organize.

5.10.4 Types of Teleconferencing

You may find the term teleconferencing misleading. Some people interpret the term to mean that television as a medium is a type of teleconferencing, but teleconferencing covers a much wider range of means which are being used in distance education or for other communication purpose. 'Tele' here means distance, i.e., talking to the students at a distance, which makes it useful for teaching at a distance. Depending on the use of hardware, there can be three types of teleconferencing:

i) Audio teleconferencing:

Teleconferencing where the audio medium is used as a two-way communication, is known as audio-teleconferencing.

Most audio-teleconferencing communication is auditory. The use of audio conferencing is rapidly becoming a preferred instructional medium in advanced countries. Two-way audio communication is a more economic method of academic interaction. For example, the University of Calgary is involved in the creation of a system using audio telephone conferencing, strictly as a teaching tool known as 'educational teleconferencing'.

There are some other universities in the developed countries using audio teleconferencing for educational purposes, but such cases are today limited in number. There have been a number of studies on the effectiveness of audio teleconferencing from the point of view of the students' learning (Ellis and Chapman, 1982). These studies show that the telephone is as effective a medium of education as is face-to-face teaching. A study conducted in Canada found that upper level undergraduate statistics students taught through teleconference did as well as or better than most campus based students and had a zero drop-out rate! Traditional correspondence instruction on the other hand reported an average drop-out rate of 40-60 per cent. Satisfaction recorded among the students was high and remote learners thought the course did not lack rigor or suitable contact.

Audio graphics

An effective variation of audio conferencing technique is the use of graphics simultaneously along with sound. The system is called as audio-graphics. A typical audio graphics set up consists of an instructor site and a number of remote sites. The instructor site consists of a speaker phone, high powered PC-based microcomputer, modem, flatbed image scanners, and high-resolution monitor. The remote sites have one or more speaker phones and a computer along with a monitor to receive images. Such a set up enables the instructor to teach in a real-time mode, showing graphics through the computer screen.

ii) Video teleconferencing:

This type of teleconferencing is arranged by combining two-way video media. This technology is in limited use in education due to its high cost and various other problems such as the linking of multiple locations by the medium of video, availability of hardware, etc. Video teleconferencing, however, has advantages over audio teleconferencing because of its visual component. Video teleconferencing increases the quality of interaction because both the teacher and expert and the student can see each other and can share their feeling and experiences. But the problem is to justify the cost involved in arranging two-way visual communication. Video teleconferencing needs, besides budgetary provision, the most sophisticated technology at both the source and receiving ends. Up linking facilities at the receiving end are required to make it a two-way interactive communication system.

Taking all these constraints into consideration, we can find a moderate way to make use of video teleconferencing, that is, we can use one-way video with audio return connection using the satellite uplink from the receiving end. This arrangement is also difficult to make in our context because of both financial and technological constraints.

iii) Computer teleconferencing:

It is the most effective way of Satellite-based Education teleconferencing. With the adequate facility of suitable hardware, information can be sent and received at the convenience of both the teacher and the student with the use of computers. Computer conferencing can be text-based or full video based. Web conferencing methods use chat and instant messaging. One can see the other person by having a webcam and streaming video. Some of the common and popular programmes that use internet teleconferencing are the Yahoo Messenger, MSN Messenger, Skype, Google Talk.

5.10.5 Designing Teleconference Sessions

As a distance educator you may be given the responsibility of organizing teleconferencing in your institution. It is therefore, very important for all of us to clearly understand the techniques of organizing teleconferences. The process of teleconferencing consists of four stages, which follow systematically one after the other.

i) Planning:

As in any business meeting, careful advance planning is of vital importance for the success of teleconferencing. First of all, decide the content of your session. Right from the initial stages, you need to be well aware of the possible issues to be involved in the discussion, and the needs and attitudes of the students participating. You need to know the number of students and the duration of each session. Check all the necessary requirements, viz., and demonstration of equipment including a trial of

microphones, etc. At the same time you need to have, on the location, discussions with the technical staff involved in the operation of teleconferencing -- every operational detail has to be discussed with them to avoid mid-conference failures.

ii) Preparing materials:

After initial planning, prepare supporting materials for the students. Educational teleconferencing should be enriched by the use of printed materials, including charts and diagrams. Printed guides should reach the students well in time so that they can use these materials during the teleconferencing session and can actively participate in the discussion.

iii) Preparing students:

Preparing the students for active participation in teleconferencing is an important function of the distance educator in charge of the conference. The students should know in advance about the content being discussed, objectives to be achieved, and about the teleconferencing system. They should be psychologically ready to learn through talk-back facilities and the preparatory activities should not take much time.

iv) Conduct in the actual session:

After preparing the students, the actual teleconferencing starts. The sessions should be interactive so that all the students can actively participate and learn as much as they can from the conference session. While conducting the session, you have to keep some do's and don'ts in mind. They are as follows: The expert should have a good audible voice and sound communication skills, and should undergo a simple audition test before being involved in a teleconference, i.e. the expert(s) should be selected carefully. The expert should speak directly into the microphone rather than to anyone side. By doing so, he/she will be able to speak directly to the students. As far as possible, each student should be addressed by name, if the number of learners is small enough. Like the self-instructional materials, the discussion should be informal. There is a need to add human touch and humor as and when required, provided it does not disturb the educational value of the discussion. It is important to set aside sufficient time for questions, answers and discussions so as to make the sessions as interactive as possible. Each student should be encouraged to actively participate in the discussion.

v) Feedback:

After the conference is over, you should co-operate with the students to satisfy their specific needs and requirements. Thereafter, they should be given sufficient time to react to the quality of the conference. At the same time the feedback session must help the students know about their performance. Almost equal time should be given for preparing the students for teleconferencing and collecting feedback from them.

For the students' convenience, the teleconference sessions could be recorded and be made available at the study centres for those individual students who could not, for any reason, participate in the conference. By doing so, you can make optimum use of the discussion held during the conference.

Towards the end of the sessions, you should motivate the students to send either personally or through letters, their reactions to and ideas about the overall effectiveness of the teleconference. The reactions thus collected can be used as inputs to improve the quality of the conferences to be held in the future. Systematic evaluation of this type provides further inputs of use to the teleconference organizers, distance teachers and experts in planning and conducting better teleconferences. The effectiveness of teleconferencing depends on the appropriateness of the content being discussed and the resourcefulness of the expert invited to conduct the discussion. An eminent scholar/expert can be brought into teleconferencing from anywhere in the country to give a live presentation to the participating student.

5.11 WHAT ARE THE BENEFITS OF USING WORD PROCESSOR IN THE CLASSROOM?

Word processors are a valuable part of the technology now available for educators in schools. Students and teachers both receive many benefits when doing classwork on the computer. Many of the headaches of paperwork are eliminated.

Spelling

Word processors contain an electronic spell checker. The student writer has immediate feedback about misspelled words. Student must discern which of the computer-generated spellings is correct for the context. Teachers no longer have to red-ink spelling errors. They can focus on the few exceptions the spellchecker does not catch.

Security

Teachers and students gain a sense of security about losing assignments. When the student saves her work, she avoids the possibility of the assignment being lost or misplaced. If an assignment is ever misplaced, a replacement can be easily printed.

Legibility

Teachers benefit by receiving a readable copy that is easy to grade. Students with poor handwriting can increase their scores with better looking papers. Students should be instructed to turn in copies of work in a readable font.

Publishing

Work done on a word processor can easily published on a bulletin board. Teachers can create electronic anthologies of their students' writings. Each student can receive an electronic copy of published works with no printing costs.

Mobility

Work done on a word processor and saved on the Internet is highly portable and accessible from any computer with Internet access. Dogs do not eat papers in cyberspace. "I forgot it at home" is irrelevant. Just log onto the nearest computer and your work appears on the screen.

5.11.1 Word Processing Software Revisited, some tips for making better use of a basic Software Application

Word processing software is installed on almost every school computer and is one of the most used technology tools in education, with applications from writing up IEP's to documenting lesson plans. Most teachers have some basic experience with word processing and students are achieving similar levels of experience. However this may not be the case for the ESOL student. Many of our students have come from situations where computer access has not been available and because of that background they are trying not only to learn a second language, but are also are trying to gain the technology skills that they see in other students. Some of the ways word processing can be used, both by the teacher and by the student, to improve language ability and learning while acquiring modern technology skills.

Consider some of the basics of word processing. First, many features of word processing can improve readability for students. The word processor makes text consistently easy to read. Whenever we write with a word processor, we're not writing for a single situation. We're writing for repeated, future uses of a document: rereading, editing, printing, etc. If a student loses instructions or assignments, it is very easy to produce an additional copy. Additionally a teacher's work is made easier when a document can be adapted for a new need rather than created from scratch. By using the word processor to write assignments, develop lesson plans, create reports, and even to write letters home, we are creating a personal database of material for future use.

An advantage of word processing is the ability to change many aspects of text appearance. Some font styles are easier to read than others are. Some font sizes are more comfortable than others are, and some font colors provide better contrast than others. Research has discovered (Tinker, 1963, Prince 1967) that very young (early elementary) or emerging readers prefer to read a san serif font (block print) and that as they develop they change their reading preference to serif fonts. In addition to using a font style that is easy to read, you can improve readability by altering the font size. Research has found that younger children read better with larger fonts, where children under seven prefer a font size of about 24 point and as the children get older the font size decreases to about 11 point. If you look in most children's books and compare their text to newspapers and books for adults, you will see the difference in font structure for children and adults. ESOL students are often times at a learning stage which mimics that of emerging readers where it is most effective for them to read san serif fonts initially before developing ability with the serif fonts. Word processors are generally set to use a specific font size. This default setting may not be the best for your situation. You can easily adjust font size by highlighting the text, going up your toolbar (or format menu) and selecting a new font size. A small increase in font size may make your written material much easier for students to understand. While the larger words will take more paper space, the document will look less crowded, therefore making students feel more comfortable with the amount of

information on the page. White (or open) space is important in documents. Too many small words filling a page can be intimidating to any student.

Another thing to keep in mind while writing is to make sure that you use both upper and lower case letters; don't write in all upper case. The shape of the written word itself produces an image to a reader that helps in decoding, by providing clues to names and sentences. Check this for yourself, by looking at a sentence written in all capital letters, then compare it to text written in both upper and lower case letters. The important consideration concerning the font style is to keep it clear and simple to read.

You can also use common word processor functions of bold, italicize and underline text to help your students recognize important words and phrases. You can build upon these functions even further with most word processors by using the highlighting option. We can also highlight words, or emphasize them, through the use of color. Most word processors will allow you to change the color of your text to emphasize a word or passage, and format the words to a new color. Many new word processors also have the ability to highlight text. In this case, the word processor acts as if you used a highlighting marker, allowing you to mark over words and passages. Research has found that the contrast that exists between yellow and black is greater than the contrast existing between black and white, making items highlighted in yellow much easier for most people spot and recognize . To make a word stand out, use the word processor's ability to highlight or color text. It is important to remember that if you have a color printer, you can print color fonts and color highlights. If you have a laser printer you may be able to use your printer to highlight in levels of gray. Using text which has levels of gray or even a slight change in color can make it very noticeable when it is printed on paper. Of course not all documents need to be printed. Many documents will be read directly from the computer's monitor, freeing you from concerns about printers, paper, or ink. Additionally students can easily adjust text that they read on the screen to a font and size that feels comfortable to them. Research has found that students' prefer a larger font size of 14-16 for reading from a computer screen.

Word processors usually include several checking features. Checkers for spelling, grammar, thesaurus and readability are useful to teachers when creating documents for student use, as well as for student authors. Rather than being a crutch or causing students to develop lazy habits, spelling and grammar checking actually provides immediate feedback and makes students more independent writers. Students can see as they write when words are misspelled. They choose from a list of options to make their own corrections. With repetition, students use correct spelling and grammar because they make the writing faster. There is no delay between writing and receiving a corrected paper, and there is no reliance on an outside authority for making corrections. An important point to teach the students here concerns the behavior of the spell checker and grammar checker. Students will usually consider anything marked by either checker to be wrong, even when this is not the case (such as their own names or passive sentence structure). You as the teacher might want to change the grammar checker's style format to a more causal setting. The thesaurus is an excellent tool which students can use to explore new vocabulary to include with their writing. Make sure to check students' writings and explain about shades of meaning, because students tend to simply add new words suggested by the thesaurus without checking their meaning. Another tool available in modern word processors is a readability statistics analyzer. Readability statistics are useful for teachers who have students reading at various grade levels. A document can be checked for readability, then adapted to higher or lower grade levels with the help of built-in readability statistics analyzer and thesaurus. Students can even track their own writing ability, providing their own feedback on their writing.

5.12 WHAT IS COMPUTER NETWORKING?

According to managed it services Toronto provider PCM Canada, Computer networking is a pool of integrated computers configured to one another. Computer networks or data networks are chains of nodes linked by communication channels. The nodes receive, transmit and exchange data between endpoints. The data transmitted is in the form of voice or video traffic.

How do you network two computers? Wired Ethernet cables or wireless radio waves connect the computers. This blend of technologies relies on defined algorithms to transmit data between

specified endpoints. The endpoints include computer servers, mobile devices, and tablets among others.Computer networking enhances seamless communication, cohesive functioning, and resource sharing.

Types of computer networks

There are several types of computer networks. The critical difference lies in the geographical areas they serve and their core purpose. These include:

Personal Area Networks (PAN)

Personal Area Networks are interconnected technology devices that serve a single individual within one facility. PAN is accessible within a distance of 10 meters and is a suitable option for small offices or residences. PAN-enabled devices include telephones, computers, video game consoles, and peripheral devices. PAN provides maximum flexibility. You can upload pictures or stream music and videos online to your device.

Local Area Network (LAN)

LANs are limited to a single building like an official establishment. LAN covers a defined area. LAN is essential for resource sharing including file servers, printers and data storage. LAN hardware such as Ethernet cables and hubs are affordable to acquire and maintain.

Small LANs manage 2 or 3 computers while large LANs host thousands of servers. Internet connectivity can either be wired or wireless. LAN is popular with many establishments due to its high speed and low set up costs.

Metropolitan Area Networks (MAN)

MAN covers a distinctly larger are than LAN. MAN consists of configured computers covering a town, city or campus. Depending on the connection, MAN can connect a single area or traverse several miles. MAN is a series of several LANs linked to create a greater network, also known as Campus Area Network (CAN) or Campus Networks.

Wide Area Networks (WAN)

WAN covers a larger geographical area than MAN. The network connects a country or even continents. And the computers are interlinked by cables, optical fibers or satellites. Most users can access the network via modems. WANs consist of smaller several networks of LANs and MANs. The internet which consists of networks and gateways is an example of a public WAN.

Home Area Network (HAN)

HAN is a computer network limited to a defined boundary like a house or home office. HAN is a type of IP based LAN which can be either wired or wireless. HAN is a broadband connection available to several users using a wired or wireless modem. It consists of shared devices like faxes, printers, scanners or data storages.

5.12.1 Importance of computer networking

There are several reasons why networking is essential to a business, institution or individual. These benefits include:

Cut back on costs

Why are networks important? You can cut back on costs and allow for efficient use of resources. Hardware is the priciest resource in technology. Computer networks reduce your hardware costs significantly.

Since the computers are interconnected, resource pooling is efficient. Printers, copiers and backup storages are shared among employees. This eliminates the need to buy single IT assets for each employee. Unlike traditional desktops, you don't need frequent software installations. You only need to install updates and track performance.

Boost storage capacity and volume

Computer networks pools their entire data to a central data storage server. This data is accessible to your employees. You can use the data to gain insights on how to boost your company's productivity. With a central server, you lower the number of storage servers needed. And you increase the efficiency of operations.

Optimize convenience and flexibility

Computer networks enable flexible operations. The data is not stored in a local server making it accessible with internet connectivity. You can access your data from any device. This enhances free movement while accessing your data wherever you may be.

Streamline communication

Computer networking is a massive boon to the communication landscape. Networking allows you to send and receive text messages and files in real time. Information is available and easy to access from any device. You only need a reliable internet connection. Even if your device shuts down, you log in from a different device and access your data.

Drawbacks of computer networking

Despite their many advantages, computer networks present a fair share of setbacks of concern to your business. These include:

Unsecured network

Data servers expose your information to man-in-the-middle attacks and viruses. Employees can download and exchange corrupt files which expose your server to viruses and malware. Shared resources open doors of opportunities for network security breaches. Hackers can easily snoop on your data.

Cost prohibitive

Installing standard networking hardware can set you back by thousands of dollars. Small equipment such as modern routers is expensive. The networks also need regular monitoring and maintenance. You will have to consult the services of a professional.

Low productivity

Computer networks are flexible, but employee productivity is a concern. The internet presents a minefield of distractions. Employees may engage in unsolicited file sharing or instant messaging during work hours. Even though most companies have drafted work regulations, policing employees is an uphill battle.

Bottom line

Quality technological solutions are a significant determinant of the success of any business. Computer networks are essential technology solutions for your business. Computer networking offers immense technological solutions to businesses and institutions in various formats. Networking pros include inexpensive setups and maximum utilization of available resources. You get to streamline communications within and outside your organization."To win in the marketplace, you must first win in the workplace"- Doug Conant.

5.12.2 Advantages of Installing a School Network

File Sharing

Network file sharing between computers gives you more flexibility rather than using floppy drives or Zip drives. You can use the network to save copies of your important data on a different computer, examples share photos, music files and documents.

Sharing Devices

Sharing devices is another benefit in which a network exceeds stand-alone computers. For example laser printers, fax machines, modems, scanners and CD-ROM players, when these peripherals are added to a network, they can be shared by many users.

Sharing Internet Access

In computer network, students can access the internet simultaneously.

Speed

Using a network is faster way for sharing and transferring files. Without a network, files are shared by copying them to floppy disk.

Cost

Networkable versions of many popular software programmes are available at considerable savings compared to buying individual licensed copies. Its allows easier upgrading of the program.

Security

Files and programmes on a network are more secure as the users do not have to worry about illegal copying of programmes. Passwords can be applied for specific directories to restrict access to unauthorized users.

Centralized Software Management

One of the greatest benefits of installing a school network is the fact that all of the software can be loaded on one computer (the server). This saves time and energy when installing updates and tracking files.

Electronic Mail

A network that provides the hardware necessary to install an e - mail system. E-mail's help in personal and professional

communication for all school personnel as it enables the spread of general information to the entire school staff. Example, electronic mail on LAN (Local Area Network) enables students to communicate with teachers and peers at their own school when connected to the Internet, it enables users to communicate with others.

Flexible Access

School networks allow students to access their files from computers throughout the school. Some schools provide public access to students to begin an assignment in the computer lab and save then access the file when they are at a cyber cafe or home.

5.12.3 Disadvantages of Installing a School Network

Expensive to Install

Large campus networks can carry hefty price tags. Cabling, network cards, routers, bridges, firewalls, wireless access points, and software can get expensive, and the installation would certainly require the services of technicians. But, with the ease of setup of home networks, a simple network with internet access can be setup for a small campus in an afternoon.

Requires Administrative Time.

Proper maintenance of a network requires considerable time and expertise. Many schools have installed a network, only to find that they did not budget for the necessary administrative support.

Servers Fail.

Although a network server is no more susceptible to failure than any other computer, when the files server "goes down" the entire network may come to a halt. Good network design practices say that critical network services (provided by servers) should be redundant on the network whenever possible.

Cables May Break.

The Topology chapter presents information about the various configurations of cables. Some of the configurations are designed to minimize the inconvenience of a broken cable; with other configurations, one broken cable can stop the entire network.

Security and compliance.

Network security is expensive. It is also very important. A school network would possibly be subject to more stringent security requirements than a similarly-sized corporate network, because of its likelihood of storing personal and confidential information of network users, the danger of which can be compounded if any network users are minors. A great deal of attention must be paid to network services to ensure all network content is appropriate for the network community it serves.

5.13 FLIPPED CLASSROOM LEARNING

The Flipped Classroom is

- A means to increase teacher contact time
- An environment that increases student responsibility
- Blending of direct instruction and constructivist learning
- A class where all students are engaged
- A class where absent students won't fall behind
- A class where all students are engaged in their learning

5.13.1 Difference between Traditional Class Room and Flipped Classroom

Traditional Classroom	Flipped Classroom
Student gets frustrated and gives up	Teacher able to assist learners when they get stuck
Teacher reviews homework in class	Students able to review their work in class with peers and teacher
Struggling students afraid to ask for help – often they don't complete assignment	Teacher able to identify students as they struggle with content and immediately provide feedback and help
Students do not read the comments placed on graded assignments	Teacher able to immediately provide feedback and help

5.13.2 Key Elements of the Flipped Classroom

- Provide an opportunity for students to gain first exposure prior to class.
- Provide an incentive for students to prepare for class. *Task associated with points*
- Provide a mechanism to assess student understanding.
- Provide in-class activities that focus on higher level cognitive activities.

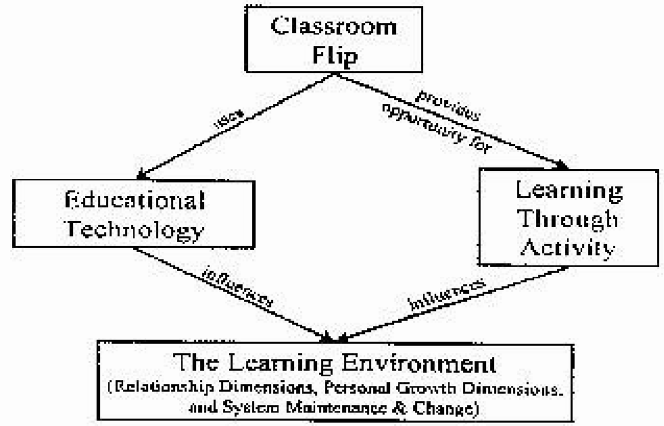


Figure 2.1. Theoretical framework

Four Pillars of F-L-I-P

- Flexible Environment
- Learning Culture
- Intentional Content
- Professional Educator

Flexible Environment

Educators can create flexible spaces in which students choose when and where they learn. Furthermore, educators who flip their classes are flexible in their expectations of student timelines for learning and in their assessments of student learning.

Learning Culture

The Flipped Learning model deliberately shifts instruction to a learner-centered approach where class time is dedicated to exploring topics in greater depth and creating rich learning opportunities. Students are actively involved in knowledge construction as they participate in and evaluate their learning in a manner that is personally meaningful.

Intentional Content

Educators continually think about how they can use the Flipped Learning model to help students develop conceptual understanding and procedural fluency. Educators use intentional content to maximize class time in order to adopt methods of student-centered, active learning strategies.

Professional Educator

Professional educators continually observe their students, providing them with feedback relevant in the moment and assessing their work. Professional educators are reflective in their practice, connect with each other to improve their instruction, accept constructive criticism and tolerate controlled chaos in their classrooms.

Roles & Responsibilities of Students and Teachers

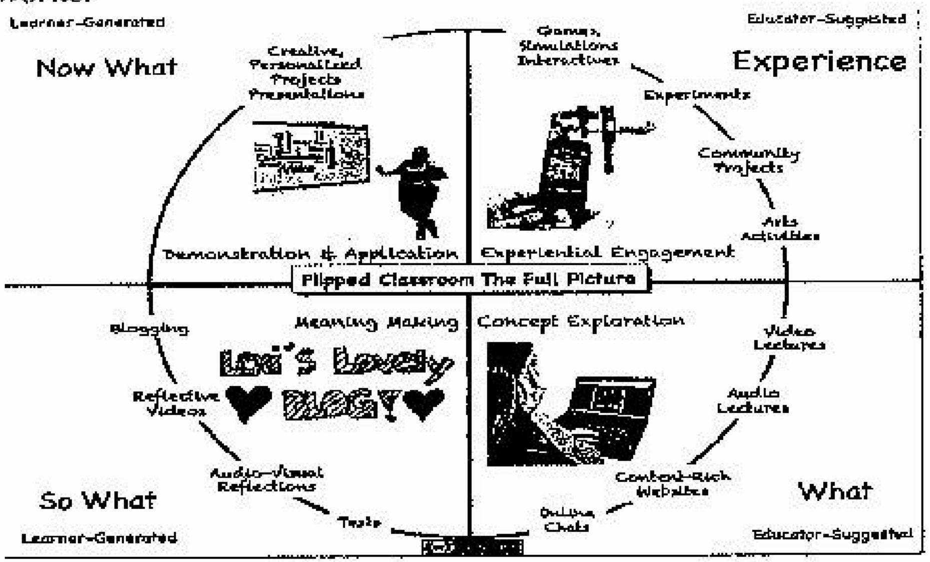
Teacher and students engaged in

- concept exploration
- making meaning of content
- Students take responsibility for their own learning

Teacher acts as Coach/ Mentor/ Guide

- Teacher helps students Access Information
- Process information
- Develop critical thinking skills needed to problem solve
- Teacher will help students to set and monitor goals

Aids in the development of skills needed by the 21st century worker



	OLD (BEFORE THE FLIP)	NEW (AFTER THE FLIP)
Before Class	Students assigned something to read	Students guided through learning module that asks and collects questions.
	Instructor prepares lecture.	Instructor prepares learning opportunities.
Beginning of Class	Students have limited information about what to expect.	Students have specific questions in mind to guide their learning
	Instructor makes general assumption about what is helpful.	Instructor can anticipate where students need the most help.
During Class	Students try to follow along.	Students practice performing the skills they are expected to learn.
	Instructor tries to get through all the material.	Instructor guides the process with feedback and mini-lectures.
After Class	Students attempt the homework, usually with delayed feedback.	Students continue applying their knowledge skills after clarification and feedback.
	Instructor grades past work.	Instructor posts any additional explanations and resources as necessary and grades higher quality work.
Office Hours	Students want confirmation about what to study.	Students are equipped to seek help where they know they need it.
	Instructor often repeats what was in lecture.	Instructor continues guiding students toward deeper understanding.

5.14 BLENDED LEARNING

Currently, **no universal definition exists** for blended learning.

Most agree that it involves some **combination of online and face to face instruction**. Blended learning does not occur simply by adding a few online strategies to a traditional classroom.

Successful blended learning requires an **intentional and integrated approach.** The blend of methods should depend upon the needs of the students and the school.

Integration

5.14.1 Implement of Blended Learning

- Blended learning allows teachers and schools to **address a variety of learning styles** with a variation of instructional methods (Wiffin, 2002).
- Blended learning practices in the K-12 environment **increase student motivation** (Berson, 1996; Lipscomb, 2003; Pye & Sullivan, 2001; Scheidet, 2003; Wellman & Flores, 2002;).
- Blended learning allows for **more personalized instruction**.
- Blended learning provides opportunities for teachers to use online curriculum for **basic information and for extensions/review** and class time for higher order thinking activities.
- Blended learning frequently **provides a higher level of interaction** than commonly experienced in face to face courses.

5.14.2 Six Models of Integration

1. Face-to-face Teacher Lead

Teacher lead course in a face to face environment with supplemental online resources

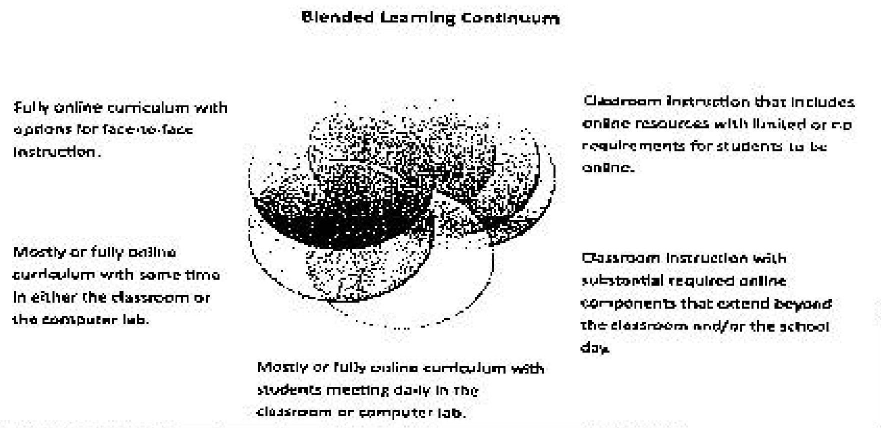
2. Flex: Most of the content is online with tutoring in a face to face classroom

3. Rotation: Students rotate between face to face and online within the same course

4. Self-blend: Students are online after hours on their own initiative

5. Online driver: The content is online, student proceeds at own pace and meet face to face occasionally with teacher and other students

6. Online lab: Students in computer lab with mentor during school day and teacher is online



5.15 ARTIFICIAL INTELLIGENCE (AI)

Artificial intelligence (AI), the ability of a digital computer or computer-controlled robot to perform tasks commonly associated with intelligent beings. The term is frequently applied to the project of developing systems endowed with the intellectual processes characteristic of humans, such as the ability to reason, discover meaning, generalize, or learn from past experience. Since the development of the digital computer in the 1940s, it has been demonstrated that computers can be programmed to carry out very complex tasks as, for example, discovering proofs for mathematical theorems or playing chess with great proficiency. Still, despite continuing advances in computer processing speed and memory capacity, there are as yet no programs that can match human flexibility over wider domains or in tasks requiring much everyday knowledge. On the other hand, some programs have attained the performance levels of human experts and professionals in performing certain specific tasks, so that artificial intelligence in this limited sense is found in applications as diverse as medical diagnosis, computer search engines, and voice or handwriting recognition.

- Artificial intelligence (AI) is a wide-ranging branch of computer science concerned with building smart machines capable of performing tasks that typically require human intelligence.

- The field of AI refers to developing computer systems that use big sets of data to perform "intelligent" tasks: think visual perception, understanding natural language, reasoning and decision making. Machine learning is one way of building such systems based on providing the computer with examples of what it should do, and let it figure out (learn) how to do it.
- AI entered the scene of computer science in the early 1950s when computers started learning checkers strategies and speaking English.
- Since then the field has made remarkable progress. With faster and more powerful computers, access to larger amounts of data and algorithm improvements, AI systems have taken off.
- Artificial intelligence (AI) makes it possible for machines to learn from experience, adjust to new inputs and perform human-like tasks. Most AI examples that you hear about today – from chess-playing computers to self-driving cars rely heavily on deep learning and natural language processing. Using these technologies, computers can be trained to accomplish specific tasks by processing large amounts of data and recognizing patterns in the data.

5.15.1 Definitions of artificial intelligence (AI)

Definitions of artificial intelligence (AI) have surfaced over the last few decades, John McCarthy offers the following definition "It is the science and engineering of making intelligent machines, especially intelligent computer programs. It is related to the similar task of using computers to understand human intelligence, but AI does not have to confine itself to methods that are biologically observable."

Stuart Russell and Peter Norvig then proceeded to publish, Artificial Intelligence: A Modern Approach (link resides outside IBM), becoming one of the leading textbooks in the study of AI. In it, they delve into four potential goals or definitions of AI, which differentiates computer systems on the basis of rationality and thinking vs. acting:

Human approach:

- Systems that think like humans
- Systems that act like humans

Ideal approach:

- Systems that think rationally

- Systems that act rationally

Alan Turing's definition would have fallen under the category of "systems that act like humans."

At its simplest form, artificial intelligence is a field, which combines computer science and robust datasets, to enable problem-solving. It also encompasses sub-fields of machine learning and deep learning, which are frequently mentioned in conjunction with artificial intelligence. These disciplines are comprised of AI algorithms which seek to create expert systems which make predictions or classifications based on input data.

5.15.2Artificial Intelligence History

The term artificial intelligence was coined in 1956, but AI has become more popular today thanks to increased data volumes, advanced algorithms, and improvements in computing power and storage.

Early AI research in the 1950s explored topics like problem solving and symbolic methods. In the 1960s, the US Department of Defense took interest in this type of work and began training computers to mimic basic human reasoning. For example, the Defense Advanced Research Projects Agency (DARPA) completed street mapping projects in the 1970s. And DARPA produced intelligent personal assistants in 2003, long before Siri, Alexa or Cortana were household names.

This early work paved the way for the automation and formal reasoning that we see in computers today, including decision support systems and smart search systems that can be designed to complement and augment human abilities.

While Hollywood movies and science fiction novels depict AI as human-like robots that take over the world, the current evolution of AI technologies isn't that scary – or quite that smart. Instead, AI has evolved to provide many specific benefits in every industry. Keep reading for modern examples of artificial intelligence in health care, retail and more.

AI has been an integral part of SAS software for years. Today we help customers in every industry capitalize on advancements in AI, and we'll continue embedding AI technologies like machine learning and deep learning in solutions across the SAS portfolio.

The idea of 'a machine that thinks' dates back to ancient Greece. But since the advent of electronic computing (and relative to some of the topics discussed in this article) important events and milestones in the evolution of artificial intelligence include the following:

- ❖ **1950:** Alan Turing publishes *Computing Machinery and Intelligence*. In the paper, Turing— famous for breaking the Nazi's ENIGMA code during WWII- -proposes to answer the question 'can machines think?' and introduces the Turing Test to determine if a computer can demonstrate the same intelligence (or the results of the same intelligence) as a human. The value of the Turing test has been debated ever since.
- ❖ **1956:** John McCarthy coins the term 'artificial intelligence' at the first-ever AI conference at Dartmouth College. (McCarthy would go on to invent the Lisp language.) Later that year, Allen Newell, J.C. Shaw, and Herbert Simon create the Logic Theorist, the first-ever running AI software program.
- ❖ **1967:** Frank Rosenblatt builds the Mark I Perceptron, the first computer based on a neural network that 'learned' though trial and error. Just a year later, Marvin Minsky and Seymour Papert publish a book titled *Perceptron's*, which becomes both the landmark work on neural networks and, at least for a while, an argument against future neural network research projects.
- ❖ **1980s:** Neural networks which use a back propagation algorithm to train itself become widely used in AI applications.
- ❖ **1997:** IBM's Deep Blue beats then world chess champion Garry Kasparov, in a chess match (and rematch).
- ❖ **2011:** IBM Watson beats champions Ken Jennings and Brad Rutter at *Jeopardy!*
- ❖ **2015:** Baidu's Minwa supercomputer uses a special kind of deep neural network called a convolutional neural network to identify and categorize images with a higher rate of accuracy than the average human.
- ❖ **2016:** DeepMind's AlphaGo program, powered by a deep neural network, beats Lee Sodol, the world champion Go player, in a five-game match. The victory is significant given the huge number of possible moves as the game progresses (over 14.5 trillion after just four moves!). Later, Google purchased DeepMind for a reported $400 million.

5.15.3 Types of artificial intelligence

Arend Hintze, an assistant professor of integrative biology and computer science and engineering at Michigan State University, explained in a 2016 that AI can be categorized into four types, beginning with the task-specific intelligent systems in wide use today and progressing to sentient systems, which do not yet exist. The categories are as follows:

- ❖ **Type 1: Reactive machines.** These AI systems have no memory and are task specific. An example is Deep Blue, the IBM chess program that beat Garry Kasparov in the 1990s. Deep Blue can identify pieces on the chessboard and make predictions, but because it has no memory, it cannot use past experiences to inform future ones.
- ❖ **Type 2: Limited memory.** These AI systems have memory, so they can use past experiences to inform future decisions. Some of the decision-making functions in self-driving cars are designed this way.
- ❖ **Type 3: Theory of mind.** Theory of mind is a psychology term. When applied to AI, it means that the system would have the social intelligence to understand emotions. This type of AI will be able to infer human intentions and predict behavior, a necessary skill for AI systems to become integral members of human teams.
- ❖ **Type 4: Self-awareness.** In this category, AI systems have a sense of self, which gives them consciousness. Machines with self-awareness understand their own current state. This type of AI does not yet exist.

5.15.4 Examples of Artificial Intelligence

- ❖ Siri, Alexa and other smart assistants
- ❖ Self-driving cars
- ❖ Robo-advisors
- ❖ Conversational bots
- ❖ Email spam filters
- ❖ Netflix's recommendations

5.15.5 Artificial Intelligence Applications

There are numerous, real-world applications of AI systems today. Below are some of the most common examples:

- ❖ **Speech Recognition:** It is also known as automatic speech recognition (ASR), computer speech recognition, or speech-to-text, and it is a capability which uses natural language processing (NLP) to process human speech into a written format. Many mobile devices incorporate speech recognition into their systems to conduct voice search—e.g. Siri– or provide more accessibility around texting.

- **Customer Service:** Online chatbots are replacing human agents along the customer journey. They answer frequently asked questions (FAQs) around topics, like shipping, or provide personalized advice, cross-selling products or suggesting sizes for users, changing the way we think about customer engagement across websites and social media platforms. Examples include messaging bots on e-commerce sites with virtual agents, messaging apps, such as Slack and Facebook Messenger, and tasks usually done by virtual assistants and voice assistants.
- **Computer Vision:** This AI technology enables computers and systems to derive meaningful information from digital images, videos and other visual inputs, and based on those inputs, it can take action. This ability to provide recommendations distinguishes it from image recognition tasks. Powered by convolutional neural networks, computer vision has applications within photo tagging in social media, radiology imaging in healthcare, and self-driving cars within the automotive industry.
- **Recommendation Engines:** Using past consumption behavior data, AI algorithms can help to discover data trends that can be used to develop more effective cross-selling strategies. This is used to make relevant add-on recommendations to customers during the checkout process for online retailers.
- **Automated stock trading:** Designed to optimize stock portfolios, AI-driven high-frequency trading platforms make thousands or even millions of trades per day without human intervention.

5.15.6 The advantages and disadvantages of artificial intelligence

Artificial neural networks and deep learning artificial intelligence technologies are quickly evolving, primarily because AI processes large amounts of data much faster and makes predictions more accurately than humanly possible.

While the huge volume of data being created on a daily basis would bury a human researcher, AI applications that use machine

learning can take that data and quickly turn it into actionable information. As of this writing, the primary disadvantage of using AI is that it is expensive to process the large amounts of data that AI programming requires.

Advantages

- Good at detail-oriented jobs;
- Reduced time for data-heavy tasks;
- Delivers consistent results; and
- AI-powered virtual agents are always available.

Disadvantages

- Expensive;
- Requires deep technical expertise;
- Limited supply of qualified workers to build AI tools;
- Only knows what it's been shown; and
- Lack of ability to generalize from one task to another.

Advantages of AI in Education:

- **Automated Grading** – Both for the objective and standardized descriptive assessments, this AI assisted procedure is been used in the developed countries and also in some universities of our country.
- **Support Teachers** – In addition to helping with grading, AI will also provide support for teachers in some of the routine tasks as well as communication with students. For example, one college professor can successfully use an AI chatbot (use machine learning and natural language processing (NLP) to deliver near human like conversational experience) to communicate with students.
- **Support Students** – It has already been suggested that in the future students will have an AI lifelong learning companion. Essentially, this next generation of students will grow up with an AI companion that knows their personal history and school history. Therefore, it will know each student's individual strengths and weaknesses.
- **Meet a Variety of Student Needs** – AI will also be able to help students with special needs by adapting materials to lead them to success. For instance, studies are already showing positive results for AI teaching Autism Spectral Disorder (ASD) student's social skills.
- **Allow Teachers to Act as Learning Motivators** – As AI takes on more of a teaching role by providing students with

basic information, it will change the role of teachers in the classroom. Teachers will move into the role of classroom facilitator or learning motivator.

- **Provide Personalized Help** – AI will also provide personalized tutoring for students outside of the classroom. When students need to reinforce skills or master ideas before an assessment, AI will be able to provide students with the additional tools they need for success.
- **Identify Weaknesses in the Classroom** – AI will also work in identifying classroom weaknesses. For instance, AI will identify when groups of students miss certain questions letting the teacher know when material needs to be retaught. In this way, AI will also hold teachers accountable and strengthen best teaching practices.
- **Monitor and Analyze Student Progress in Real-Time** – Teachers can monitor and analyze students' progress in real-time using AI tools. It means that the teachers do not need to wait until they compile annual report sheets. Also, AI gives teachers' recommendations as to the areas that require repeat or further explanation. In this instance, AI smart analytics picks up on topics that most of the students struggled with.
- **Saves Time and Improves Efficiency** – AI does is handle the burden of repetitive tasks teachers and schools have to deal with daily like attendance. There are also custom writing services like 'Online Writers Rating' that help handle any repetitive writing tasks. It helps save up more time so teachers can focus on teaching the students and other essential duties involving creative work. For example, when using a grammar tool, the teacher doesn't have to repeatedly correct students' grammar. The students can use the AI-powered tools to learn word pronunciations, meaning, and proper usage. AI education is also beneficial to international students that are still struggling to learn a new language.
- **More Personalized Learning Experience** – AI in education enables schools to carve out personalized learning experiences for their students. From student data, AI can analyze the student's learning speed and needs. With the results, schools can personalize course outlines that enhance learning based on students' strengths and weaknesses. For instance, they can include topics that appropriately suit learning requirements.

- **Convenient and Improved Student-Teacher Interactions** – AI education makes interaction more comfortable and convenient for both students and teachers. Some students may not be bold enough to ask questions in class. Such could be as a result of the fear of receiving critical feedback. So with AI communication tools, they can feel comfortable asking questions without the crowd. While on the part of the teacher, they can give detailed feedback to the student. Sometimes, there isn't enough time during classes to respond to questions in detail. They can also provide one on one motivation for any student that needs help.
- **Simplifying Administrative Tasks** – Every educational institution has tons of school admin tasks they need to deal with daily. Including AI to their systems can help to automate such tasks. It means that administrators can have more time to run and organize the school more smoothly.

5.15.7 Challenges of Artificial Intelligence in Education:

- **Cost of AI Technology** – AI education comes at a high price. As new technology emerges, budgets will have to increase to cover the expenses. Besides the installation of AI software, schools will also need to consider the cost of maintenance of the software.
- **Vulnerable to Cyber Attacks** – Artificial intelligence software is highly vulnerable to cyber-attacks. Considering that it contains a ton of data, hackers are constantly devising ways to attack.
- **Little to no Room for Flexibility** – No matter how analytical AI robotics can be, it cannot flexibly develop a student's mind as a teacher would. While educators can proffer multiple problem-solving methods, AI doesn't have alternative teaching methods.
- **AI also operates on garbage in garbage out basis** -While it might detect errors, it cannot correct them. So whereby there's a human error in imputing information, AI still carries out the analytic process. But the final result will read that there are errors. Hence time is wasted, and the process has to be repeated all over again.

5.15.8 Uses of AI in education

Microsoft and McKinsey's recent report of over 2,000 students and 2,000 teachers from Canada, Singapore, the UK, and

America shows that artificial intelligence (AI) is already providing teachers and schools with innovative ways to understand how their students are progressing, as well as allowing for a fast, personalized, targeted curation of content.

- **Personalized learning:** Managing a class of 30 students makes personalized learning nearly impossible. However, AI can provide a level of differentiation that customizes learning specifically to an individual student's weaknesses and strengths.
- **Teacher's aid:** Teachers don't only teach, they also spend hours grading papers, and preparing upcoming lessons. However, certain tasks, such as marking papers, could be done by robots, giving teachers a lighter workload and more flexibility to focus on other things. Machines can already grade multiple-choice tests, and are close to being able to assess hand-written answers. There is also potential for AI to improve enrollment and admissions processes.
- **Teaching the teacher:** Artificial intelligence makes comprehensive information available to teachers any time of day. They can use this information to continue educating themselves in things such as learning foreign languages or mastering complex programming techniques.
- **Connecting everyone:** Because AI is computer-based, it can be connected to different classrooms all over the world, fostering greater cooperation, communication, and collaboration among schools and nations.

5.16 AUGMENTED REALITY

5.16.1 Using Augmented Reality in the Classroom

Augmented reality has quickly become one of the most popular trends among software and hardware developers. Augmented reality allows people to interact with their real-world environment in an enhanced way. Using different kinds of technology, the environment is augmented by the technology. This can occur in many ways. On the more complex end, users can get feedback through their sense of touch, their sense of smell, and their sense of sight and sound.

The technology being used changes the user's perception of the real world. This is not virtual reality, because virtual reality attempts to replace the real-world environment with a digital one. Depending on how sophisticated the technology, augmented

reality can display visuals over the real world, provide artificial audio feedback that's not actually present in the environment, and give various types of feedback through a user's different senses.

If this sounds complicated, that's because it can be. Advanced augmented systems include visual headsets and complicated controls for interacting with the enhanced real-world environment. However, there are much simpler ways of implementing augmented reality. This simpler version of augmented reality can be accessed using the same tools that most students use every day – the modern cell phone. Smart phones, specifically, have the ability to create an augmented reality experience.

As documented by Forbes, there are multiple ways that augmented reality has been brought to life over mobile phones. The Gatwick app connects terminal beacons to mobile phones to help passengers navigate the area. Meanwhile, IKEA's app lets people scan their rooms and redesign them on their mobile screens using Ikea furniture. Sephora's app lets people scan their faces and try different looks by combining different facial products. These are just a few examples of how reality can be enhanced using augmented reality apps. However, how can augmented reality help in the classroom?

In a piece developed for EdTech magazine, author Meghan Cortez noted that augmented reality had the power to transform the classroom. The possibilities, Cortez argued, were vast. Augmented reality was being integrated into medical training to help medical students simulate surgeries. In primary and secondary schools, augment reality was used to help deaf students by delivering interactive flashcards that operated using sign language to interact with the students.

Teachers should be aware that the application of augmented reality has, thus far, leaned heavily toward science, humanities, and the arts. These are topics where augmented reality lends itself most easily toward interactive experiences. On the other hand, the least integrated topics were in areas of health, welfare, agriculture, and teaching training. This divide suggested that augmented reality might not be ideal for teachers of all subjects. However, it can provide an important, hands-on experience that helps to increase the level of engagement students feel with their materials.

With all this discussion about the tools that teacher's use and the benefits of augmented reality, how are teachers actually putting the technology to use in the class? Writing for Education Week Teacher, author Kate Stoltzfus listed approaches teachers should use to integrate augmented reality into the classroom. This may help you integrate augmented reality into your classroom more easily.

One of the primary ways that teachers can ease the transition to augmented reality is by adopting tools that can apply to all grade levels and skill levels. A specific example that educators used was Merge Cube. This cube is essentially a hologram that students can manipulate. Depending on the apps that are downloaded, the cube can become many things – a globe students can use to explore the world, a multi sided game that trains students in ways that sharpens their memory, an aquarium that students can use to identify various fish species, or a device for rotating projections of the solar system. Merge Cube is just one example of a single tool that can be adapted for multiple audiences using an abundance of free apps. Tools that can be adapted to multiple audiences can help reduce the amount of resources teachers have to seek out to enhance their classes.

Another way that teachers are integrating augmented reality into their classes is by giving students ownership of the education experience. Teachers often use augmented reality to deliver content through apps. However, students can put together their own projects using apps that can then be shared with the class.

Whereas students used to put together PowerPoint presentations, they can now use augmented reality to bring presentations to life. Just one example that educators provided was the use of Google Street View. By drawing on images from Google Street View, students created images that could be viewed at any angle. This helped them create views of the world that could be manipulated in 360 degrees. Such approaches helped to increase student engagement and get them doing hands on work in augmented reality.

Other ways that teachers have integrated augmented reality was discussed at New Gen Apps, an organization that develops mobile applications for business and education. One interesting way to help get students acquainted with their instructors can be accomplished using augmented reality. Schools can put together a faculty photo wall where students can use their devices to scan each photo. Then, on their screens, the photos will come to life and provide different information about their teachers. This can be

especially helpful the first week of school as students are first getting acquainted with their instructors. Using this approach, a novel and fun way of learning about a school's staff may help students transition into the school year.

Science teachers have also used augmented reality to provide students with a more engaging way of learning about lab safety. Using triggers around the classroom, science instructors can bring warning about lab devices to life. As students walk through the lab, their devices can scan these tags and learn about different rules they should adhere to when running different lab experiments and working with different equipment.

It's clear that there are many interesting ways that teachers are using augmented reality in the class. It can be used to teach rules and safety protocols, introduce students to their instructors, empower students to create their own three dimensional presentations, or manipulate adaptable tools that can teach different lessons depending on the app that's downloaded. However, because augmented reality remains a relatively new field in education, researchers and educators are sure to discover new ways of using the technology in the future.

The state of augmented reality in the classroom today is still a little bit fractured. Researchers have made significant strides in understanding some of the best approaches to integrating augmented reality, but it remains one of the newest fields in education. Because of that, educators are still finding new ways of effectively using the technology in the classroom.

Still, like all things in education, augmented reality is a tool that can be best put to practice with a little planning. Creating lessons that use augmented reality to enhance, rather than replace, more traditional instructional approaches seems to be the best approach to using the technology. This can be done by using the many apps that make augmented reality such a flexible and powerful tool for use in the classroom. The technology also presents an exciting way for teachers to refresh their own instructional practices as well

5.16.2 Benefits of Augmented Reality

Augmented reality can promote interactive experiences with coursework, encourage collaboration between students, improve motivation, and increase learning gains. These benefits all rely on effectively implementing augmented reality into the class. When integrated poorly, augmented reality faces several limits, including too much focus on virtual information and intrusion of the technology onto actual leering gains.

There are other benefits to adding augmented reality to the classroom as well. For instance, as covered by the Tech Advocate, there are plenty of ready-made resources that teachers can draw upon to more easily integrate the technology. Several books have already been released that are designed to work with augmented reality, and the only thing that students need to bring to the lessons are their smartphones.

Teachers only have to provide the books and students can scan the book pages, bringing them to life on their phones. This means less prep times for teachers. With regard to resources, the availability of resources means that teachers only have to print out enhanced worksheets if they want their students to take the augmented experience home. Pre-made augmented reality sheets can be printed out by the teacher, which makes it easier for teachers to make augmented reality into their preparations.

The staff of Cypher Learning, which develops the NEO learning management system, also reviewed some of the benefits of using augmented reality in the class. When properly integrated, it can increase enthusiasm for the materials among students and increase their motivation. Augmented reality often presents a new way of learning for students that's easy for them to become enthusiastic about.

At the heart of what drives this increased motivation is the novelty of augmented reality, but also it's interactivity. Students can directly manipulate enhanced elements of their environment, which can contribute to deeper learning among students. It's more likely that students will remember their materials for the long term because they will be able to directly interact with their materials in fun and innovative ways.

5.16.3 Designing an Augmented Classroom

When it comes to implementing augmented reality technology into the classroom, there are a few design principles that instructors should keep in mind. As discussed by researchers Robert Miller and Tonia Dousay, the first duty of a teacher is to ask *why*. Why are they implementing the technology and how do they intend to use it to support learning goals? Haphazardly

introducing a technology just for the sake of using it won't lead to improved outcomes.

The best role for augmented reality is as a supplementary tool, rather than as a primary method of delivering content. So, while teachers can use it to bring lessons to life, they also need to maintain a focus on more traditional ways of delivering content. Augmented reality is best suited to enhancing those traditional lessons rather than acting as stand-alone methods of instruction.

It's also important for teachers to know their classrooms. If their students don't possess a certain level of technological knowhow, they won't be able to work with augmented reality effectively. If teachers are hoping to use such technology, they should probably gauge their classroom at the start of the year first. This is important because Miller and Dousay suggested that trying to use this technology among students who didn't have a certain level of technological competence would only detract from meeting learning goals.

5.16.4 Augmented Reality Apps

One of the difficult aspects of integrating augmented reality into the classroom is settling on the right apps to use. There are more than just a few apps teachers can draw upon to enhance work in the classroom. Here are just some of the most popular for students of all ages.

- **Popar Toys.** This app changes the way that students engage with stories and puzzles. The books and puzzles that can be used with the app come to life with animation and read longs as the children make their way through the stories. This app brings stories to life, which can help children become more interested in a range of topics. They can even engage with interactive books about the world and the solar system, helping bring science to life.
- **Augment Education.** Augment Education is targeted at an older crowd than Popar Toys and is best used by high scholars. Students can use this tool to create presentations in augmented reality and create three dimensional designs. The ability to make models in augmented reality can bring lessons to life in classes that range from health to architecture. Teachers can also create lessons in 3D, making Augment Education one of the most innovative and helpful apps for educators.

- **SkyMap.** One of the most interesting augmented reality apps is Google's Sky Map. Specialized specifically for classrooms, this app helps students learn more about the solar system. The interesting part about this app is just how hands-on it is. When students direct their devices upward and to the sky, the app finds planets and stars that are in that direction. The app's ability to find stellar bodies lets students identify different parts of their universe. Besides showing various constellations, the app challenges students to find certain planets and displays high quality images of interesting locations like the Orion Nebula.
- **Fetch Lunch Rush.** Yet another app that has appeal for a younger audience, this PBS produced app used printable cards to aid the augmented reality experience. With this game, kids are asked to feed a movie crew, but they have to solve math problems to do so. Kids spread out their cards across the game board and kids have to move from piece to piece, using the cards to bring to life unique problems for them to solve.
- **Human Anatomy Atlas.** One of the most intriguing augmented reality apps for education is the Human Anatomy Atlas. The atlas bring to life three dimensional representations of the human body that students can interact with. Besides providing models that students can interact with, the app also includes descriptions of different types of disease and injuries and even questions that students can use to test their knowledge. This is yet another app that's fantastic at the high school level and is obviously perfectly suited for health classes.

These apps are just some examples of how educators can put augmented reality to use in a variety of topics. Augmented reality is a powerful tool because of its flexibility. It can be targeted not only at different age groups but at different education fields as well.

5.17 COMMUNITY RESOURCES

The main aim of using the community resources is to give equal opportunity to all the students to take part in such activities and to enrich their interest and understanding the contributions made by other streams to the teaching of commerce. Moreover, it should always be executed through the active and willing cooperation of the students, staff and school management.

5.17.1 Meaning

A community provides 'concrete', 'saleable' and 'tangible' resources which are extremely 'dynamic' 'interesting and 'meaningful' for the teaching and learning of commerce. It is not enough for a child to have knowledge about factories, farms, council sessions, museums and commercial enterprises etc. He must have the acquaintance with all these. A community is a child s laboratory for having first hand learning about the ways of living Community resources are the people and places members of a given community can turn to for assistance in filling an unmet need. The organizations can be public or private.

5.17.2 Types

Commerce Association or Forum:

Commerce Association or forum may be organized in the school under the leadership of the commerce teacher. He should take necessary initiative steps to run it. It should have elected president secretaries, cashiers, executive members and general body members; it should conduct the various activities of the department, i.e. other co-curricular activities.

- It provides a valuable link between students and staff and makes a worthwhile contribution to the smooth running of the departments.
- It provides students with useful practical experiences in real situations.
- It increase the relationship among students,
- It provides opportunity to the students to take various responsibilities.
- It may conduct the meeting with former students to know about their work and experiences in business.
- It can arrange the guest lectures with the great businessmen and professional men to know about their jobs and services.
- It can conduct film sessions related to work of committee and subjects.
- It is the mother organisation for all activities related to commerce. All the office bearers appointed by the students are responsible for the organisation of the programme and thereby they gain useful business experience.

Exhibitions:

Organisation of the programme and thereby they gain useful business the commerce department can conduct exhibitions on important occasions. They can exhibit charts, diagrams, graphs, models, pictures and scrap book collection of coins and rupees used at various periods.

Debates and competitions:

The commerce association or department in the school can organize debates in commerce subjects. It can also organize elocution and essay competitions on various topics in commerce and in general. Students of other school inside and outside the town can also be invited. It can also organize annual competitions in short hand and typewriting- Prizes and certificates be given to outstanding students. Because of these competitions students may get interest in studying the subject matter. It eliminates the stage fear of the students. It provides an opportunity for improving vocabulary and expression.

Commerce Magazine:

The commerce association can also publish a school or its department magazine annually. The commerce teacher can ask the students and teachers to contribute articles to the magazine. This department, if possible can publish monthly or weekly written magazines.

Social service:

The commerce teacher can also organize social service scheme, this can be in the form of laying village road, cleaning the temple, teaching the adult illiterates, conducting evening special classes to the school children with the help of commerce students.

Vacation work:

The commerce teacher can make an arrangement with the hostel warden and business agencies to engage the commerce students in their accounts departments during vacations. This will train the students with the variety of experiences during the limited period. Students with this experience can discuss with the other students to transmit their experience.

The commerce association may also make the necessary arrangements for guest lectures, panel discussions, workshops etc. He can invite nearby college or university professors, bank managers, great businessmen, auditors and accountants etc., to give guest lectures. He can also utilize the help of the above mentioned people to conduct a Workshop, panel discussions etc. In addition to the above mentioned activities the commerce teacher can organize other co -curricular activities according to the

needs, interest, and co-operation available from the students and others.

Mock "Job Application Interview"

Mock interviews on job applications may be organized in the schools. These may be the practical applications of theoretical details furnished by the commerce teacher as how to apply for some posts and how to face interviews. This activity may be organized for students in the form of drama. Dramatic performances will have far-reaching effects on the mind of students. They will remove the shyness and fears of interview mania and will enables students to farewell in the interview which they may face later on.

For the success of these activities, interested and talented students may be selected and involved in inviting applications for vacancies of posts of assistants, clerks, stenographers, accountants; salesmen etc. Other students, who are not involved in the activity directly, should apply for various posts. All applications should be scrutinized and various types of questions should be framed, which may be asked in the interview by the members of the interview board. Students should also be trained efficiently how to reply to different questions put to the candidates in interview in order to make the function more attractive, real and life like proper dress for the students participating in the activity may be arranged from the professional firms dealing in different materials used in drama and other functions.

One of the commerce teachers or a businessman may be requested to act as the chairman of the interview board. At the first instance, interview can be taken in private or in the presence of a group of students, where the 'deficiency may be pointed out for improvement. Thereafter they should be analyzed and weakness of students may be removed or overcome before the students apply for actual posts later on in their life.

5.17.3 Uses in Teaching and Learning in Commerce

The following are the few uses of using community resources.

a. It gives through knowledge about the subject matter.
b. It develops a sense of individual and collective responsibility among the students.
c. It develops the ability to work with others
d. It utilizes the leisure time in s better way.

e. It helps the students to become a good citizen
f. It creates and maintains good teacher student relationship
g. It makes school life more attractive and encouraging.
h. It develops good relationship between school and home,
i. It gives experience in leadership and organisation.
j. It strengthen the knowledge acquired in the classroom
k. It brings out the inner potentialities of the students.
l. It develops the self-confidence among the students.
m. It develops the feeling of belonging adjustment adoptability in all situations.

As a whole it develops the all-round personality of the students and a spirit of service;
The students learn how to organize, plan conduct the various community related activities.

5.17.4 Establishing Link among School, Resources and Community

The potential activities in school/community linkages fall into three main categories.

1. Community as a Source of Student Curriculum Based Learning

The community offers many opportunities to add value to curriculum based learning. There are many examples where community activities and issues have been incorporated into the school curriculums.

2. Incorporating Learning into School Contribution to Communities

Students contributing to their local community can foster learning. For example, community volunteering program allows students to experience community service and gain skills. Schools could offer student skills to community members such as in computer training. Community events involving the school such as eisteddfods or festivals create a community profile for schools and build the experience of students. Community involvement can also build student interaction with particular groups in the community such as senior citizens, indigenous people or people from a non-English speaking background. It can build the involvement of students in young people's organisation.

This interaction not only enriches the educational experience of students but also redefines students and the school as key community assets as young citizens, not just adults in the making.

3. Community Involvement in the School

Community members can add to the life of the school and to educational outcomes. Several schools have established community mentors that participate in classroom interaction and school activities.

Schools have also been seen as centers for community learning where community members and organizations can assess school facilities for meetings and learning activities.

4. The Role of Teachers

Community linkage can help orient new teacher .community situations can also be used in professional development of teachers. Teacher may also play a role inassisting community organisation or in helping the local community manage local issues activities.

School is a social institution set up by the society to serve its ends. It is charged by the society with the duty of training and bringing up the students so that they may be able to take part effectively, harmoniously and efficiently in the group to which they belong. With the emergence of progressive education, a new doctrine came into being that the surrounding community is the best teacher. It presents lively and interesting material for learning. In the words of Clark and Starr: "Extending the classroom into the community can make a course exciting and forceful, for every community is a gold mine of resources for teaching."

Whatever experiences a child gets in the community are very important from educational viewpoint. He gets real experiences in the community. Bookish knowledge of government, finance, factories, rivers, dams, agriculture, crops etc. is not enough. In the capacity of an able citizen, he should possess knowledge of all these in practical terms. Several vague economic problems become clear when seen in practical context in the community, because they are present in their concrete form. Factories, shops, office etc. located in the community are effective means of acquiring knowledge. Students become aware of their actual circumstances and can resolve their doubts by speaking to those working in them

A Commerce teacher should make a file for all those community resources which he can mobilize for learning in Commerce and Accountancy. These resources can be used in Commerce teaching in the following two ways:

- Taking school to community and
- Taking community to school

Let us see elaborately one by one

5.17.5 Taking School to Community:

We can take school community in order to make use of different teaching aids available there and provide students an opportunity to come into direct touch with them. There are several such resources such as field trip, survey, community service, etc.

(i) Field Visits

Field trips help in real learning. Students learn about real objects on teaching spot, therefore, their learning becomes real and lifelong. These are very important in Commerce teaching. Students can be taken on a visit to industries or commercial centres and give them real time experiences on industries, trades, banks, insurance company, share market, jewelry market, mandis etc.

While undertaking a field visit, the following points should be kept in mind by a teacher:

1. A teacher must accompany students.
2. A teacher should know about the place of visit and related environment.
3. He should give necessary instructions to students.
4. Students should be asked to carry diaries to note down any points they observe.
5. A critical discussion should be held after the trip.
6. Students acquired knowledge should be evaluated.
7. A field trip should be planned well.
8. Approval of officials, means of transport and meals should be arranged well.
9. The places of visit should be selected informally.
10. All formalities should be carried during and after the field trips, such as thanking the concerned people, submission of report to Principal etc.

(ii) Social Survey

A community survey is a study of the status of something in the community. We divide a survey into two parts;

(a) The survey based on concrete and objective data and

(b) The survey based on opinion of the people.

A survey should be well-planned, so that students can be acquainted with the realities prevailing in the society. An unplanned survey cannot meet its objectives. A survey based on concrete and objective data is more accurate and real. The scope of survey is quite wide in Commerce teaching.

(iii) Community service schemes

The school can be taken to community with some community service schemes. They may be study of different community aspects such as those belonging to provision of water, fodder, medicine, beautification of village cleanliness of wells, adult literacy scheme, woman literacy scheme etc.

A community service scheme should be selected carefully. A scheme should be simple and easy and should not need large resources, time or money, so that it can be undertaken easily and which students can complete by their efforts. It should be in such a manner that is appreciated by the community.

(iv) Museum

The objects gathered by teachers and students are stored at one place and exhibited in the form of a museum. Classifying it into different section. Such objects can include currencies and coins of different countries, rocks, artifacts, wood, types of iron, postal stamps, magazines, pictures, idols, statues, images etc. A teacher should encourage students to gather as many objects as they can do and thus enrich the museum.

(v) Exhibition

An exhibition too is an important aid in commerce teaching. Students can be taken to exhibitions to learn about many aspects. They are valuable in commerce teaching. Different types of objects are displayed in an exhibition. Students watch them, satisfy their curiosity and thus enrich their knowledge. It also helps in gaining latest knowledge about different dimensions. Exhibitions on means of transport, means of communication, agricultural improvements, crops, minerals etc. are very valuable.

A teacher should be careful while taking students on an exhibition. He should do it with due planning. Besides taking students to an exhibition, a teacher can also arrange an exhibition himself. A commerce teacher can contribute in it greatly. Students can be asked to make models and specimens of coins, charts, maps etc. Making of these objects too helps in learning. When students watch these objects, they come to know about different things. A teacher should assign students the duty of displaying objects before the visitors. A discussion should be held at the end of an exhibition.

(vi) Excursion

An excursion or field trip is a trip arranged by the school and undertaken for educational purposes in which students go to places where the materials of instruction may be observed and studied directly in their functional settings. The commerce teacher can make an excursion to the factory, market, industries, industrial towns, places, multipurpose projects, conducting, and marketing survey in the villages, important trade centres and agricultural farms and conducting population survey. It is an important aid in teaching commerce that establishes some concepts and learning the firsthand experience to the usual context of classroom teaching. The commerce teacher must make careful planning of excursion if the experience is to be educationally worthwhile. The students should be encouraged to have definite ideas about what they want to see. They should also be encouraged to write a report about the whole excursion

5.17.6 Taking Community to school:

Lectures by members of the community are an important method of taking community to school. Other means can be teacher-guardian unions, fairs, local festivals, national festivals etc.

Lectures

Lectures can be invited from prominent people of the community in order to acquaint students with the actual conditions in the community. These people can introduce students with different aspects of the community. From the viewpoint of economics teaching, these people can belong to different occupations, such as ironsmith, goldsmith, weaver, banker, industrialist, etc. They can tell students how to succeed in their occupations and other aspects. Secondly, teachers from other

colleges and universities can be invited for lectures. These lectures are very important.

5.18 COMMERCE AND ACCOUNTANCY CLUB

Teaching is a science as well as art. Teaching commerce and making it interesting for students is a challenging job for commerce teacher. Commerce subject is related to all national/ international happening in economic, political, and social areas. So, it is very essential for a teacher to connect her /his teaching to practical life. It is equally essential for a teacher to provide learning experiences beyond the four walls of the school for achieving this, Commerce and Accountancy Club would be a great help.

The club is an organization where different activities take place to facilitate the actual teaching-learning process. It is something more beyond textbook teaching and normal classroom activities.

Each and every individual has got his own creativity, attitude, and interest which could be explored through the different activities.

Commerce and Accountancy Club is more informal and it takes care of the various domains of learning. Club activities are more practical. There is no assessment for club activities while club activities represent freedom of expression in classroom teaching represents conformity and repression.

5.18.1 The organisation of a club

- **Decide what kind of a club it will be** - the people in the club have a common goal and share a common interest.
- **Figure out where and when your club will meet** - the activities should be organized once or twice in a week at a venue with timing mention by members.
- **Being recruiting members for your club** - selection of pupils for the post of president, vice president, Treasure, Secretary, Advertising team for the smooth activities of the club.
- **Meeting has to be organized once or twice in a week**
- The teacher discusses with the head of the institution describing the importance of Commerce and Accountancy Club in the modern age. We will be discussing the goals and purpose of the club.

- **Decide the rules and procedures of the club.** Members will be deciding the rules and procedures of the club i.e, communication procedure, discussion, various club activities, preparation of newsletter and magazine, etc. For the club, there should be an inspiring slogan.
- **Keep moving forward.**

5.18.2 Activities of a club

1) Dramatization, Role play/ Street play

In this play, members can do the role play/street play on a few topics of commerce like OC subject for better clarity and in-depth knowledge. Some of the topics are responsibilities towards an employee, customers, investors, etc.

2) Teaching aids preparation

Every person is unique and they have different talents. This club will help the students to prepare teaching aids like chart papers, cutouts, videos, PPT, etc. This will create interest among students and topic can be understood easily and they find the up to date knowledge in that topic.

3) Field visits, tour, etc.

Commerce is a totally practical oriented subject. So, students should have practical knowledge through field visits, tours, etc. Students will come across the current scenario around the world. This may create more information among the students rather than in a textbook.

4) Project work, Survey

Project is very important aspects for the students. They should get the freedom to select the topic as well as teacher should give them those topics which may help them in the future. This will create interest among the students towards the subject.

5) Celebrations of days

Students can celebrate different days in club-like World Consumer Right Day – 15th March, National Technology Day – 11th May, World Press Freedom Day – 3rd May, Telecommunication Day – 17th May, Labour Day – 1st May, etc.

6) Talks, session by experts

In this club, they can organize talks and sessions by experts in different fields. So, a student gets a deeper knowledge of the content as well as they will have a different angle of looking the same thing/ topic.

7) Seminars, workshop, conferences

Students can take in seminars, conferences, and workshops on different topics. Their knowledge will extend with a different perspective of peoples who are presenting. They will knowledge of other subjects and how can we connect with other subjects.

8) Display boards, bulletin boards

Every student should get a chance to display boards and bulletin board. So, that they will bring cutouts from magazines, newspaper, journals, etc. They will help in their up to date knowledge and what happening around the world. They can categories boards according to the topic.

9) Competitions

Members, as well as students, can organize competitions for students like essay writing, speech, debate, etc. They will learn by themselves with their interest according to the topic and competitions. This may result in students will motive for further studies.

10) Banking/ Cooperative stores.

In this club, students can run the bank as well as cooperative stores. Students will get the real-life experiences of it. This may develop different skills like decision making, problem-solving, different values, etc.

11) Discussion / paper reading

12) Exhibitions/ fare

Above are a few activities which can be conducted in Commerce and Accountancy Club.

5.19 COMMERCE AND ACCOUNTANCY RESOURCE CENTRE – FIELD VISIT

5.19.1 Field Visit

A visit to a place that is made by the students to learn about something. It is done in the natural setting. It is total a practical approach. It gives a better understanding of the students towards the concepts. It connects classroom life to practical life or real life. Commerce is totally based on practical life. if a teacher cannot connect the topic with real life then it does not make any sense or it's not sufficient lecture. The student develops interest by themselves in the field.

5.19.2 Definition of Field Visit

A visit (as to a factory, farm or museum) made (as by student and teacher) for purpose of first-hand observation.

A visit to a place that gives the student the chance to study something in an aerial environment, rather than in a classroom or a laboratory.

Where a student can go for a visit

Bank, export-import, marketing, cooperative society, retailer shop, different departments in banks, Agro Industries, consumer form, Stock Exchange, etc.

5.19.3 Field Visit Procedure

- ❖ Field visit/excursion if properly organized could be very useful in teaching some concept to students.
- ❖ There should be adequate preparation for the visit and notes should be carefully made of things observed during the visit which the teacher must have earlier on described to the students in class. The teacher should visit the site before the date of their excursion and make adequate arrangements.
- ❖ The teacher should choose place to go for field visit with education value.
- ❖ The place chosen must be safe.
- ❖ The concepts to be learnt must have relevance to their core curriculum.
- ❖ As soon as the visit is over, the teacher should guide the entire class in evaluating the results so that maximum educational value can be derived from the visit.
- ❖ The students should be allowed to ask questions on their observations.
- ❖ Finally, students must write a report of the visit.

5.19.4 Organisation of a visit

- ❖ purpose of the visit
- ❖ Finalizing of the place and venue
- ❖ the safety of the students
- ❖ instruction to the students
- ❖ obtain appropriate prior approval

Planning

The planning is the first step of organisation of a visit. It will include place, permission, date, timings, schedule, background information, objectives, transport optional/ Arrangements,

student instruction related to visit, divide the class into groups, schedule of the day, parents' permission letter.

So while planning we have to take care of all the things which are listed above.

Implementation

Implementation is a second step of the field visit, where actual students are going for the Visit. So, the students already know what day will be doing, they will talk to the people, they will find a function, challenges, talk to the people, documentation, etc.

Evaluation

Now we will discuss, how was the Visit, student experience, report on the Visit, and also regarding the social values of the skills that developed during the Visit. This can be done both in the class as well as in a report.

5.19.5 Importance of field visit

The following are the importance of field visit

1. **Develop social skills**

   Through field visit, students will develop social skills like communication, self-discipline, Team Spirit, Corporation, coordination, tolerance and understanding the diversity, etc.

2. **Connecting practical experience to school or practical life**

   It will connect the classroom environment with the practical life of a student. It may help the students for better understanding of the concepts by connecting the topic to their real life.

3. **Making commerce subject more interesting**

   Through field visit, we make the commerce subject more interesting by giving practical knowledge.

4. **A better understanding of the concept or topic**

   There are few topics in the textbook which lead to practical knowledge. So, that topic can be taught through a field visit. They get a better understanding of the topic.

5. **All types of learners**

   Field visit hence all types of learners. it is a process which includes every kind of learner. this makes the field visit more meaningful and helpful for the students.

6. **Supplement to the classroom teaching**

   A field visit is a supplement to the classroom teaching which gives practical experience to the students and there is a change in the teaching method where students learn by themselves or through visits.

7. **Information of learning environment and hand on experience**

In this visit, students learn in an interactive way by communicating with other person and with hand on activities done by them.

8. **Apply the lesson in the Real world**
9. **Real world experience**

Students will get real-world experiences related to the field there are going. They will know the challenges faced by them and what exactly the world needs from them which may improve them.

10. **Uncommon opportunities through innovation exploration and exposure.**

Above are a few importance of field visit.

Example: bank visit

EDUCATIONAL IMPLICATIONS OF BANK

- To develop knowledge of various financial institutions, relevant to economics
- To develop an understanding of the different activities, departmental coordination at the banking system
- To develop an understanding of correlating the status of any economy and natural resources.
- To promote an understanding of taking an intelligent decision in future [with respect to finance and resources], decision making, risk bearing,
- To enhance the capacity building for self-generated employment.
- It helps the students and the teachers to promote human resources.
- Economists develop a comparative study of the economy of various nations & verify the defects in the Indian economy.

5.19.6 Importance of Field Visit in Teaching

Educational Field Visits can associate with a variety of benefits for teachers and students. In terms of teaching, it would be better for the teacher to take part in field Visits mostly because of its advantages. It is true that the Visits provide teachers with ample opportunity to develop personality and professionalism.

Obviously, it is necessary for teachers to get in touch with innovative practices and the latest teaching methods to apply these ways into the curriculum at school. Educational field Visits help teachers widen their horizons of knowledge and broadening the scope of their syllabus. Most educational Visits pay much attention to outdoor activities with a wide range of aspects in life. It means the Visits allow the teacher to connect the classroom with real-life and authentic experience. Therefore, it is considered a fantastic way to improve their lessons and catch up with innovative teaching which is more practical and meaningful. Most importantly, the teacher is a guide to inspire and stimulate students to understand the world that exists beyond the school curriculums. To be more precise, students will be encouraged to learn not only knowledge but also soft skills, life skills, communication skills and more.

Other benefits of educational Visits are that they give teachers the chance to meet up experienced and minded teachers. This is a chance for them to exchange and share the experience as well as teaching methods. It also allows them to widen their circle friends and learn from fellow colleagues. Obviously, educational field Visits are effective methods to teach difficult subjects such as history, biology, or physic. It is easy for students to learn these topics by seeing and experiencing real life. For example, teachers can bring the students to a historical museum in order to learn stories, events, and characters in history subject. With a field Visit on a biology class, teachers can guide your students to observe certain types of insects trees or animals. It is true that the method is a fantastic way to help students get a deeper understanding of biology subjects through real life. Not only that, but learning in educational Visits also helps students to have so much fun. With support, help, and guide from teachers, educational tours can act as a mobile class for students. This is the reason why the numerous benefits of an educational field Visit for teachers cannot be overlooked.

5.19.7 Importance of Field Visit in Learning

In terms of learning, an educational field Visit is one of the most fantastic ways for young students to learn through real life. It is true that the educational field Visit is a perfect combination of sightseeing, learning, traveling and hands-on learning opportunities. These Visits contribute significantly to the cognitive development of children mostly because educational travel allows students to witness authentic things beyond the theory of subjects by actually experiencing and seeing in real life. Obviously, there is no better way to learn something than to see or do it for yourself. Students will develop a sense of enjoyment and feel the subject closer without the pressure of doing exercises or being called to answer a question. It erases the boredom of classroom lectures and gives students the opportunity to visit new places and new environment which are good ways to awaken students' interest and learn by actually doing a hands-on experience. On educational tours, students can interact with other people from all sections of society. It gives them a chance to gain new perspectives, and learn from complete strangers with different opinions. Along with professional skill development, students can get a chance to learn local customs and build independence and confidence. Most importantly, field Visits will often create a lot of learning styles and make them excellent teaching tools for students. It is not simple as lectures in the classroom, field Visits apply proactive learning such as seeing, observing and practicing. It is apparent that educational field Visits play an indispensable in learning. Therefore, the mode of Visits should be encouraged and developed.

5.19.8 Importance of Field Visit in Student Life

It would not be completed your Visits if you forgot to set up an educational field Visit in your student life. It is time for you to learn new things and get a deeper understanding of life. Having an educational Visit in the early part of the term is so wise since it will allow students to bond with classmates they may not know very well. Getting away from school a day is always exciting for students and gives students an opportunity to spend time with each other in a new environment. Most of the educational tours organized with groups so it is beneficial for students to develop a sense of community. They can chat, observe and learn about each other. Taking students into a new as well as unfamiliar environment provides them with an amazing experience of traveling with peers and sharing responsibilities. Most importantly, the educational Visit provides valuable lessons away from the classroom, without using textbooks and other tools used in a normal school curriculum. To be more precise, if students have a chance to visit the science center or historical museum, they will be excited to learn these subjects with authentic experience by seeing the historic events from his/her own eyes.

Not only can that, during the course of the journey, students save unforgettable memories in their student life which is indispensable to anyone. No matter what memory is, it definitely becomes a necessary part of your student life simply because it gives them ample opportunity to learn from their own experiences and from the experience of others. Because of these reasons below, parents or schools should encourage educational tours in schools and colleges to develop the personalities of students. The benefits of educational field Visits contribute greatly to career development and social skills development later in life.

5.19.9 Importance of Educational Tour

It is true that there are various benefits to take from educational excursions. Though the primary purpose of educational tours is to educate students, they are also used as part of the curriculum to cover a wide range of life skills including teamwork, time management, communication, etc. This is because educational travel allows students to dip in fresh experiences, and use them to enhance their career prospects. If you are setting up an educational tour, you should spend the time to research useful information and prepare the necessary things for this. For example, you should pay much attention to the means of transportation, the budget, the number of students, etc. The importance of educational tours for students can be better in various ways. First of all, this tour provides ample opportunity to exchange and learn many skills with each other. On educational field Visits, students feel the sense of enjoyment which offers them to acquire a fresh perspective, learn new things and witness different sides of the country. This opens up endless possibilities to understand the world. Going on educational tours means to have a major educational element including giving students the chance to build closer bonds with their classmates, experience a new environment and enjoy a day away from the classroom. Obviously, it is highly beneficial for them to develop a sense of community and enhance their communication skills. Most importantly, it also helps the teacher meet certain objectives of the curriculum by designing tours suitable for specific requirements. Teachers need to choose the kind of experiences for a student, which depends on ages, purpose or time. One of the most effective ways is to consult a professional educational travel company that can help you to have the best tour. Obviously, whether you are a student, a teacher or a curious parent, you should set up an educational tour for young people to save unforgettable memories as well as learn amazing things.

5.19.10 Advantages of Field Visit

(i) Field Visit helps students to use their sense of observation.
(ii) It helps students to relate what is taught in the class reality.
(iii) It aids the development of enquiring mind among them.
(iv) It will also help in retention of knowledge for longer period.

5.19.11 Disadvantages of Field Visit

(i) It is time consuming
(ii) It can be dangerous if not properly planned.

5.20 QUALITIES OF A GOOD COMMERCE TEXTBOOK

A teacher gets print resources in different forms. The textbook is the most familiar print resource for the teachers and the students. It is developed on the basis of guidelines provided in the curriculum. A case study of the development of textbook by National Council of Educational Research and Training (NCERT) is given below. It provided guidelines for teaching-learning facts, principles, theories and processes in science.

5.20.1 Development of NCERT textbooks of Commerce at upper primary stage:

The textbooks were developed based on the recommendations of NCF-2005. Keeping in mind learner's existing knowledge and experiences, the textbooks introduced simple activities to explain the associated concepts.

The present science textbooks focus on activities, experiential learning and teamwork. These textbooks emphasize on learning by doing. Some guidelines for developing science textbooks are as follows:

- Make use of children's own experiences, or start the lesson with some observations or situations familiar to children.
- Do not attempt to start from the 'beginning'. These are usually more difficult at this stage and may prevent children from learning useful things. For example, children can learn the useful skill of reading a thermometer without knowing its construction, or how it works.
- Introduce chemical changes and chemical equations without teaching symbols of elements and their valences. A common

symbol H2O, which children may come across in other contexts, is explained simply as another name for water. They are told that they will learn the 'language' of chemistry in later classes.

- Do not introduce abstract concepts too early in their lives as it makes them shun thinking and resort to rote learning.
- Lay little stress on formal definitions. For example, it has been thought sufficient at this stage to define acids and bases through their tastes, and not through a pH scale. When there is little understanding of the underlying concept, children tend to cram the definitions without understanding their content.
- List 'simple' activities which can be performed without a laboratory or a kit by using mostly things lying around and discarded. Having done something herself, the child is not likely to forget it soon. She also feels more confident of her abilities to perform activities. Choose examples from child's own environment, and from within her experience. For example, in the Chapter on *Pollution of Air and Water*, make use of children's own experience the basis to compare the quality of air in places such as parks, busy roads, residential complexes and industrial areas.
- Have language which is simple and direct, so that children can themselves read and understand; they can use the books even when the teacher is not present.
- Provide opportunities to learn by cooperation and collaboration, by role play, and from peers.
- Minimize information overload by transferring much of it to the *non-evaluative* boxes. In a box in Chapter *Sound*, (NCERT Textbook of Science Class VIII, Section 13.5) for example, a Table gives the loudness of sound from various sources. This is useful information, but children do not need to cram it.
- Let teachers have the freedom to replace the suggested activities by those that they consider more interesting and suitable in their situation. They can develop their own activities, too. This flexibility aims at unleashing the creative potential of teachers.
- Attempt to help children learn the real-life skills for protecting themselves, their families and community at the time of natural disasters like floods, storms, cyclones, lightning and earthquakes.

- Attempt to sensitize children to issues related to environment, health and hygiene, gender, water scarcity, energy conservation, differently-abled persons, superstitions, prejudices, myths and taboos and their social responsibilities now and in future as citizens of India.
- Connect life outside classroom by means of non-evaluative boxes, case studies, extended learning and exercises highlighting social issues.
- Pay special attention to the development of skills like reading scales, data presentation in the form of tables, and making graphs; and habit of inquiry. Include interesting stories, facts, anecdotes and case studies, so that children enjoy going through the books. Issue cautions, wherever necessary, to prevent accidents in the classroom and outside during the performance of activities.
- Design varied types and varied difficulty levels of 'Exercises' at the end of chapters. They should require application of learning to unfamiliar situations. These should also include the open-ended questions, which are aimed at stimulating thinking, enhancing the power of expression, and discouraging rote-learning.
- Provide addresses of websites where material for further reading, better illustrations and animations are available.
- Use better printing and colour illustrations to make books attractive to children.

5.20.2 Criteria for Evaluation of a Textbook

Following criteria may be considered for evaluation of a textbook:

1. **Correctness of the content:** Is the textual content of the topic accurate, authentic and up-to-date
2. **Science as an integrated subject:** Are the concepts of science integrated with environmental component and social issues?
3. **Age appropriateness of content:** Is the content appropriate for the age level of the learners?
4. **Appropriateness of language:** (i) Is the language appropriate and effective for the learners of that particular class? Is it easy to understand? Is the language correct (spelling, grammar, etc.) and style (vocabulary, sentence structure, etc.) simple?
5. **Representing ideas:** Does the material include appropriate representations of the key ideas?

6. **Taking account of students existing ideas:** Does the material take into account of learners' existing ideas related with the concept?
7. **Introducing terms meaningfully:** Does the material introduce technical terms only in conjunction with experiences of learners? Does it facilitate thinking, promote inquiry and effective communication?
8. **Promoting students thinking about phenomena, experience and knowledge:** Does the material include tasks and questions to promote learners' thinking and reasoning about observations and experiences with phenomena?
9. **Providing relevant variety of experience with phenomena:**
   Does the material provide relevant multiple and varied experiences with phenomena to support the key concepts?
10. **Encouraging curiosity and inquiry:** Does the material help teacher to create a classroom environment that welcome learners' curiosity and encourages inquiry?
11. **Engaging students with relevant context:** Does the material provide relevant context of learners' environment?
12. **Justifying activity sequence:** Does the material include a logical or strategic sequence of activities (versus just a collection of activities)?
13. **Providing practice:** Does the material provide tasks or questions for learners to practice skills or use knowledge in a variety of situations?
14. **Assessing for understanding:** Does the material asses understanding of key ideas and avoid allowing leaners a trivial way out, such as repeating a memorized term or phrase from the text without understanding?

5.20.3 Popular Commerce Books

- Over the past few decades a large number of voluntary organizations have been established that engage themselves in bringing out popular science books in print form.
- These books try to explain the involved scientific concepts in a simple manner through examples, illustrations and anecdotes. Such a material is now available in the country in different regional languages. A science teacher can make use of these books to learn and explain scientific concepts effectively.

5.21 QUALITIES OF COMMERCE TEACHER

Following are the important qualities for a successful commerce and accountancy teacher.

❖ **Scholarship:**

Scholarship in Commerce and Accountancy is the minimum requirement to start with. This means a sound knowledge of the subject matter. Today, the concept of teaching is fastly changing. Now it is more child centered than subject centered. That is, we teach pupils, with subject as medium in order to obtain the desirable changes in pupils; the teacher must be well equipped in the subject that will aid to produce those changes. He must have a thorough knowledge of Commerce and Accountancy at the higher secondary level. Most of the teachers might have specialized only in special fields such as income tax law and practice and transport and cooperation in their B.Com and M.Com courses. They may not have a broad outline or idea of commerce and accountancy as it is required at higher secondary school level. Therefore they should enrich the knowledge in addition to what they had already acquired by going through the subject matter in commerce and accountancy at the higher secondary school level.

❖ **Professional training:**

The second essential requirement to become a successful commerce teacher is professional training. Teaching is a profession and the teacher must be perennial student. The commerce teacher must have up to date knowledge and through understanding of the present banking system, commerce, industry etc., since he has to teach all these ever changing aspects to the students. This cannot be neglected if one wants to be a successful teacher. There are various ways by which a teacher can equip himself in service. Some of the important ways are;

- ❖ Reading professional journals like, ICA, ICWA, ICS, Business journals, SEBI, etc.,
- ❖ Getting further education through distance education/ Open University course and evening/part time college course.
- ❖ Attending orientation and refresher courses through any institutions,
- ❖ Updating experiences through using electronic media, computer related coursers, field travel etc.,

Reading is a means of self-improvement for the teacher. Fast and vast reading helps him to find out the best books and

magazines in his field. Number of periodicals and magazines are available for his improvement and some of them are, The Indian Economic Journal, Cooperation, Management Accounting, Indian Journal of Marketing, Finance and Development, RBI Bulletin, Journal of Indian institute of Bankers, etc.,

❖ **Personality:**

His personality is the third essential requirement for the commerce teacher to be successful in his profession. It is very difficult to list comprehensively and analyze all the qualities which enrich his personality. For the purpose of convenience we can divide all the elements, related to personality into three division namely physical aspects, passive virtues and executive abilities.

❖ Physical aspects:

It is very important because it creates an impression and respect. It deals with those aspects which give us our first impression of individuals. Apart from other qualities, the important are,

1. ***Personal appearance***: It includes dress, facial expressions, mannerisms and personal cleanliness.
2. ***Recognition of the amenities of life***: It includes good manners, observances of social form courtesy and refinement.
3. ***Voice***: It is very important because of its effect on children is more and it is related to successful teaching. If one has a shrill, monotonous voice it is his duty to try correct as far as possible.
4. ***Good language***: It includes pronunciation, enunciation and grammar. Fluency of language makes the teaching very effective and interesting.
5. ***Health***: It affects many of the qualities and traits of teacher. Good health is conductive to the cheerful disposition and to the intelligent optimism.

❖ Passive Virtues:

The passive virtues include those qualities which make a teacher a power in the lives of his pupils. The most important in this list are,

a) ***Friendliness:***This implies having goodwill toward one's pupils and a deep interest in their welfare.

b) ***Sympathy and Understanding:*** This means that one must realize the feelings of pupils and must appreciate and understand their problems and difficulties.

c) ***Sincerity:*** It makes the teacher to dedicate his life to the profession. He must be loyal and sincere in his work.

d) ***Tact:*** The teacher must do the right thing at the right time under the most trying circumstances.

e) ***Fairness:*** The teacher must behave with all students fairly without any distinctions or favoritism.

f) ***Self-Control:*** The teacher must control his emotional feeling. This means keeping a level head under all circumstances.

g) ***Optimism;*** The teacher must have optimistic approach. There is no place for pessimist in the training of children.

h) ***Enthusiasm:*** The teacher must be enthusiastic about life about people and about his subject.

i) ***Patience:*** Teaching is a trying profession and calls for endurance and perseverance. It requires forbearance towards the weakness and faults of others.

Executive Abilities:

Executive abilities are those abilities which are found in leaders. They are absolutely essential for a teacher because he is a leader for his students.

A. ***Self Confidence and Self-Reliance***: The teacher must have a confidence in himself and his ability to carry out the objectives of education which he has set for his pupils.

B. ***Initiative:*** He must take initiative in all activities. He must lead in all fields because teacher is primarily a leader not a follower.

C. ***Adaptability and Resourcefulness***: The teacher must be able to adopt and carry out his plan in any circumstance. He must have the ability to meet any situation to carry out his plan.

D. ***Organizing Ability***: The teacher must have the organizing ability. It is very easy to plan but putting the plan into workable condition is very difficult.

E. ***Directive Ability***: Planning and organizing alone is not enough to get success. He must have the directing ability. He must be able to give direction to others.

F. ***Industry:*** The teacher must work hard just like leaders. Hard work is a planned way is very important quality of the teacher.

Other Qualities:

In addition to the above qualities mentioned, the commerce teacher should possess the following additional qualities;

1. Good Character.

He should be a responsible teacher and must have a sense of duty. His good conduct and character will have impact on the character of children, as the human mind is always imitative. Teachers are the hero-worships of their students. No teacher can inculcate discipline and good character among students if he does not deserve this himself. He should cite examples of his personal integrity, moral character and discipline.

2.Aptitude in the 'teaching profession':

If a person, joins a teaching profession after having failed in getting suitable employment has no real interest for the teaching profession, and then it is bound to have an adverse effect on the mind of students. Due to insufficient development of the topic on the part of the teacher, students may think that this subject is a dry and difficult one. The Mudaliar commission stated, "Teachers must develop a new orientation towards their work. They should take up on their work as a great social and intellectual adventure." The Commerce and accountancy teacher's relations with the students should be polite, and friendly. He should show sympathy to his students.

3. Good natured, happy and well adjusted.

4. Thorough in his subject:

The Commerce and accountancy teacher should be thoroughly grounded in theory and practice of his subject matter and possessed of the knowledge and skills necessary for teaching theory and practice in an integrated manner.

5. Use a variety of effective teaching learning procedures.

6. Trained by using various techniques:

The Commerce and accountancy teacher should be trained in the use of variety of instructional techniques such as committee of work: question and answer and demonstration, project and discussion etc., He should be competent to develop, construct and use a wide variety of teaching aids. He should have a good understanding of the psychological principle common to all learning, as well as of the psychology of skill building, basic factors in skill development and learning difficulties peculiar to commerce courses.

7. Able to develop and use instructional materials:

The teacher of commerce and accountancy should be able to use different audio- visual materials to make his instruction more effective and meaningful. He should be fully conversant with the ways in which students are prepared for receiving benefits of the teaching aids to the maximum extent. He should be able to prepare and use simple and inexpensive charts and models to make this instruction more effective.

8. Organize subject matter for instructional purposes.

9. Appreciate the value of learning about the business occupation in the area served by the school. He should also be able to play ways of, enriching syllabi, adopting programmes to community, providing necessary background to those who need and wish to continue higher education, identifying sources of supplementary teaching materials, preparing good lesson plans needed to develop knowledge, understanding skill and attitude for a wide range to student's abilities.

10. Use variety of methods to evaluate pupil's progress.

The commerce and accountancy teacher should understand the values and limitations of various methods of evaluating students' progress. This should include, understanding values and limitations of various types of test, knowing how to prepare, administer, improve and refine teacher made tests, knowing how to utilize rest result as a basis for remedial teaching, understanding and testing the skill development concepts and using tests and rating scales to evaluate skill performance.

11. Use appropriate equipment and machines.

The Commerce and accountancy teacher should be able to handle different types of office machines effectively; particularly those machines on which instruction is give in the school. Apart from that, the teacher should also be familiar with reference books general and commercial educational periodicals.

12. Conducting carrier guidance programmes:

The commerce and accountancy teacher should be able to discuss the vocational implications of the commerce programmes and interpret the job situations to students and parents: to prepare information materials on the objectives of the commerce

programmes: to contribute to the cumulative records maintained by the school: and to interpret the role of the class room teacher in the placement programme of the school.

13. Organize and supervise the co-curricular activities:

The Commerce and accountancy teacher should recognize the need, importance and value of co-curricular activities to the total educational programme of the school and have skill in organizing and managing activities, such as commerce clubs, field trips, cooperative stores, surveys of former students, school publications, exhibitions economic surveys, etc.,

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QUESTION FOR DISCUSSION AND RECAPITULATIONS

1. Explain Classification of Instructional Media.
2. Describe Use of Mass Media in Classroom Instruction.
3. Explain Emerging Media and their types.
4. Explain about Satellites and their types.
5. What are the Relationship between Mainstream Media and Emerging Media?
6. What are the Similarities between the Mainstream Media and Emerging Media?
7. What are Differences between Mainstream Media and Emerging Media?
8. Explain Benefits of Emerging Media for Journalists.
9. Narrate the New Emerging Media and types of emerging media.
10. Spell out the SATELLITES and EDUSAT.
11. Explain Teleconference.
12. Tell about Word Processor, What are the Benefits of Using Word Processor in the Classroom.
13. What is Computer Networking?
14. Explain Flipped Classroom Learning.
15. What are Difference between Traditional Class Room and Flipped Classroom?
16. Explain Blended Learning.
17. Explain the Artificial intelligence (AI).
18. Describe Augmented Reality.

MODEL QUESTION PAPER
PEDAGOGY OF COMMERCE AND ACCOUNTANCY
PART -I

Time: 3 Hrs **Marks: 70**

Section – A (5 x1=5 Marks)
Objective Type Questions
Answer ALL the Questions.
Each Question Carries 1 mark

1. What are the aims of teaching Commerce and accountancy at school?
   a) To nurture the natural curiosity, Aesthetic sense and creativity
   b) To develop good Character
   c) To develop the power of observation
   d) All the above

   (Ans: d)

2. When Bloom's Taxonomy of educational objectives was proposed
   a) 1956 b) 1957 c) 1958 d) 1961

   (Ans:a)

3. The methods and Practice especially as an academic subject or theoretical concept
   a) Analysis b) Subject- expert c) Pedagogy d) Expertise

   (Ans: c)

4. Which of the following is true about Project Method?
   a) It is impractical
   b) It promotes the coordination of the physical and mental activities of the child
   c) It makes the learner passive
   d) It is teacher centered activity

   (Ans: b)

5. Which of the following is a teaching aid
   a) Working model of Electric bulb
   b) TV
   c) OHP projector
   d) All of the above

   (Ans:d)

Section- B
Short Answer Type Questions.
(Maximum of 250 Words or Two and half pages for each questions)Answer any THREE questions.

6. Explain Meaning, Nature and Scope of commerce and accountancy.
7. What are the Meaning, Concept and cycle of Micro Teaching?
8. What are the steps Preparing Lesson plan in commerce and accountancy?
9. Explain Team Teaching and their advantage and disadvantage
10. What are the uses of mass media in classroom Instruction?

Section -C
Essay Type Questions.
(Maximum of 500 Words or 5 pages for each questions)
Each Question carries 10 marks

11. a) Describe Instructional objectives and behavioral objectives of commerce and Accountancy

    Or

    b) Explain the processes of Bloom's Taxonomy of educational objectives

12 a) Practicing a Link Lesson with Multiple Teaching Skills (For 20 Minutes)

   Or

   b) Explain suitable any 3 Skills for your subject

13 a) What are Levels of Teaching- Memory, Understanding & Reflective Level?

   Or

   b) Explain Herbartian approach to lesson planning

14 a) Describe Characteristics, Types and steps of team teaching and their advantage and disadvantage

   Or

   b) Spell out Activity Based Learning (ABL) and Active Learning Method (ALM)

15 a) Tell about Word Processor, What are the Benefits of Using Word Processor in the Classroom?

   Or

   b) Narrate the New Emerging Media and types of emerging media